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AUTOMOBILE MANAGEMENT SYSTEM

ABSTRACT

In Today's world with so many cars, models in the market, it is hard to find out which car has a high maintenance cost/index that is authentic source.Problem statement in Description o System that shows the health and Maintenance Index of various components of car models or car parts based on multiple factors.This will help new buyers to understand the maintenance costs of a certain model and probability of which car part requires more often servicing /change, OEM's to understand which part is requiring frequent change and needs to be recalled and made better in the new models. Vehicle maintenance patterns across car models by mileage, usage , age of the vehicle , regional patterns across dealers, service stations and car manufacturer

Practical and reasons why this idea could be a challenge from Implementation o Data Challenge : Build a dealer/service station network , OEM, consumers who can feed data into the system –without data this solution will not work o Ability to integrate this solution easily with the current systems which can collate the data o Marketing challenge : Owner within the Govt to take this ahead and invest this product o Political : OEM's might not want such a system to be developed.

Domain Bucket o Transport , Vehicles • Technology Bucket o Integration across systems – Integration platform across systems with security model for data extraction o Analytics & Data Science (if possible) -Bigdata/Hadoop, AI / ML o Data Aggregation o Visualization

Nowadays vehicles are important in our day to day life. Our transportation methods mainly depend on vehicles, as per the need

increases the care for the vehicle also increases, but in this busy life we fail to give much attention to the vehicle. This results in the major mechanical problems in the vehicle. Due to the busy schedule, people fail to give servicing to the car at the proper time. People forget to check the battery voltage and engine temperature. To avoid such problems our system helps the user to take good care of his vehicle by monitoring the vehicle parameters like fuel level, engine temperature and battery voltage. It also checks whether the driver is drunk or not. It gives the servicing alerts to the driver.

CHAPTER-1

INTRODUCTION

More than two –third of automotive customers indicate that Service Convenience is a determining in selecting a brand or purchasing from a Specific dealership. Consequently, Customer Service and service management is of vital relevance to ensure ongoing Customer loyalty and retention and ultimately, dealer Profitability. Dealer Business Management enables efficient Service order Processing and billing designed for any given number of orders per day

Motivation

.it includes Service requests and Scheduling, Optimization of techicians, tools, and parts as well as their deployment and scheduling

Scope

.User friendly interfaces provides ease of use and the integration technology ensure a seamless and smooth business process into OEM channel System, including ,job and Package Catalogs, Vehicle history files ,Warranty Systems, etc. Service monitoring and analysis increases the Visibility in Overall fixed Operations and helps increases service Capacity utilization, efficiency and decreased operating cost. Recreation Vehicle service technician inspect, test, service, and replace every system installed in a recreation Vehicle with the exception of the dry.

Outline

Both intervals are equally important for properly marinating your Vehicle Remember all Toyota dealerships offer a broad range of Parts and Service. The Driver and Vehicle Licensing Agency, the Driving Standards Agency and the Vehicle and Operator Service Agency Provide Services for 42 million drivers. The Vehicle and Operator Services Agency (VOSA) Provides a range of licensing, testing and enforcement services with the aim of improving the roadworthiness standards of Vehicles ensuring the Compliance of operators and drivers, and Supporting the independent Traffic **Commissioner**

This is a software which can be used by a automobile center for keeping the records of items which is to be transacted for purchase and sell. It helps to keep a record of all the

items which is to be transacted for purchase and sell. It helps to keep a record of all the transactions and a password login is created for security.transactions and a password login is created for security. In proposed system we do not have to maintain everything manually. Throught In proposed system we do not have to maintain everything manually. Throught this system if any transaction occurs it is corresponding entries is done automatically this system if any transaction occurs it is corresponding entries is done automatically because database management system gives facility of having relationship between the because database management system gives facility of having relationship between thetable.table.In proposed system we do not have to maintain different accounting books.In proposed system we do not have to maintain different accounting books. Advantages of proposed System Advantages of proposed System Easy to generate report for any transaction Easy to generate report for any transaction. It is very much faster than manual system. It is very much faster than manual system .Easy and fastest record finding technique.n Easy and fastest record finding technique.It is very much flexible to work.It is very much

CHAPTER-2

LITERATURE SURVEY

TITLE: Supporting decision-making for solving design issues in the development phase of automotive vehicles

AUTHOR: Timothé M. Sissoko

ABSTRACT: - The purpose of this research is to support decisionmaking for solving design issues in the development phase of complex systems supported by numerical simulation. We conducted our studies in a multinational car manufacturer. The first part of the research was devoted to identifying the difficulties encountered in the issue resolution process, with a particular focus on decision-making issues, methods and tools. A qualitative study done with 11 experts and on 40 decision problems highlighted that the decision-makers choose from a set of process alternatives rather than artifact alternatives. The consequences of these process alternatives such as recalculating, integrating information, waiting for the technical definition of the vehicle to evolve, etc. are not explicit. We identified the lack of a rigorous framework as an opportunity for improvement. The second part was therefore to propose a framework to support design decisions. Concurrent engineering, resources constraints and project management issues have been often overlooked in the Decision Based Design literature. Attempting to bridge this gap, we designed IRDS framework. Through IRDS, we propose to make explicit the process alternatives, to gather economic data and expert forecasts in adecision model based on the prescriptive decision theory, including the maximization of the expected utility and the economic value of imperfect information.

The third part of the research is related to the impact of uncertainty on the data collection process and on the overall decision outcomes. This has been done through proposing a sensitivity analysis that is performed with available data, before data gathering through elicitation process. The impacts on the decision-making process and information exchanges between stakeholders, as well as the resources consumed by the new practices we proposed have also been studied on a more superficial level. This work was inparticular deployed and tested on 5 cases studies. The validation of this approach requires to collect

further empirical evidence to support the hypothesis that better decisions are made on the long run. We are confident that our research will serve as a base for future studies on the design and the implementation of frameworks addressing industrial challenges.

TITLE: design and implementation of vehicle maintenance and services system for hallmark automobile services

AUTHORS: Mario Collotta

ABSTRACT: -The vehicle maintenance and service system is an automation of the various vehicle services needed by a vehicle user in a mobile application. This application provides vehicle service reminder while providing vehicle related solution. The vehicle maintenance and service system provide repair cost estimates to help mitigate the rigged system vehicle users pass through for vehicle repairs and maintenance. This project work contains a review of existing systems related to the proposed system and the prototype development methodology used to develop the system. And also provides the design of the system using the Unified Modelling Language as well as testing of the system. The examination of the literature uncovered a number of empirical evidence areas, and the resulting discussion provides the basis for the design and implementation of vehicle maintenance and services system.

TITLE: Predictive Maintenance in the Automotive Sector

AUTHOR: Marianna Ruggieri

ABSTRACT: -With the rapid advancement of sensor and network technology, there has been a notable increase in the availability of condition-monitoring data such as vibration, temperature, pressure, voltage, and other electrical and mechanical parameters. With the introduction of big data, it is possible to prevent potential failures and estimate the remaining useful life of the

equipment by developing advanced mathematical models and artificial intelligence (AI) techniques. These approaches allow taking maintenance actions quickly and appropriately. In this scenario, this paper presents a systematic literature review of statistical inference approaches, stochastic methods, and AI techniques for predictive maintenance in the

automotive sector. It provides a summary on these approaches, their main results, challenges, and opportunities, and it supports new research works for vehicle predictive maintenance. Sunil Auto Sales Pvt. Ltd. Is an Automobile selling company situated at Alambagh Lucknow. The company has dealership of Bajaj Automobiles. The company has also a service point, for providing all types of servicing of bajaj vehicles, also deals with spare parts and accessories of its vehicle.Company has mainly concerned with three departments: Sales, Service and Parts.Sales department deals with the selling of new vehicles of bajaj auto. In this department various type of tasks performed at the time of selling a vehicle which are as follows,Complete detail of the customer such as Name, Address, City.

TITLE: Management System Design for Automobile Maintenance and Repair Based on Strusts2

AUTHOR: Quan-Peng Ji

ABSTRACT: This article analyzes the current management problems of auto repair companies, and faces the needs of traditional auto repair companies for the development of informatization. Based on the standardization of the actual business process and the efficient management mode of the company, it integrates the current information system development process and combines the MVC design model. Design the data model, function model and structure model of the automobile maintenance and repair management system. Designed with Struts2 technology, which is mainly used in web service systems. According to the design idea of modularization of functions, simple and friendly interface, the system structure is divided into: business processing layer, decision-making management layer, and system maintenance layer. From the system architecture, it is divided into: application display layer, database layer, data access layer, and business layer. It can meet the basic needs of convenient and efficient management of automobile maintenance enterprises, and also leave a good impression on customers, thereby improving the management efficiency and income of automobile maintenance enterprises. Since there are still many influencing factors in the operation process of automobile maintenance enterprises, the main goal of further research is to increase and improve the

functions of the system and conduct overall management of automobile maintenance enterprises.

TITLE: Design and implementation of vehicle management system based on the ubiquitous network

AUTHOR: Ling Wu; Fei Qiao; Jianfeng Lu

ABSTRACT: With the rapid increase of automobile holdings in the domestic, how to effectively manage vehicles in regional range and improve the effective utilization of limited resources are problems that cannot be ignored by the team managers. In order to solve the above problem, this thesis takes “Peace travel” ubiquitous network demonstration system as the background, based on the technology of Flex, proposes one kind of vehicle management system which can run cross operating system operation, accessible, easy maintenance, running in the Flash Player.

CHAPTER-3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Existing system is a manual one in which users are maintaining books to store the information like product details, Distributors details, purchases, sales details and accounts for every month. It is very difficult to maintain historical data.The following are the disadvantages of the existing system. It is difficult to maintain important information in books. More manual hours need to generate required reports. It is tedious to manage historical data which needs much space to keep all the previous years' ledgers, books etc. Daily sales and purchases details must be entered into books are very difficult to maintain.

3.1.1 DISADVANTAGES

- Increased Repair Costs: When vehicles are not maintained regularly, small issues can become bigger problems that require expensive repairs. Over time, the cost of these repairs can add up, leading to increased expenses and reduced profits

- Decreased Safety: Vehicles that are not properly maintained are more likely to experience mechanical failures or breakdowns while on the road. This can put the safety of drivers and passengers at risk, as well as the safety of other drivers on the road.

- Reduced Vehicle Lifespan: Regular maintenance can help extend the lifespan of a vehicle. When vehicles are not maintained properly, they may require replacement sooner than expected, which can be a significant expense.

Current system is a manual one in which employees has to submit their applications for transport facility as well as for cab facility. Employees has to follow up regularly with transport personnel to know the status of their requests which is time consuming and hectic.

The following are the disadvantages of current system

1. It is difficult to track the occupancy.

2. More manual hours need to generate required reports

3. It is tedious to track the details of cabs provided by third party.

4. There is no possibility to track the approvals of requests.

5. No co-ordination between various departments.

3.2 PROPOSED SYSTEM

Proposed system is a software application which avoids more manual hours that need to spend in record keeping and generating reports. This application keeps the data in a centralized way which is available to all the users simultaneously. It is very easy to manage historical data in database. No specific training is required for the employees to use this application. They can easily use the tool that decreases manual hours spending for normal things and hence increases the performance.

3.2.1 ADVANTAGES

The following are the advantages of proposed system

1. Easy to process requests

2. Can generate required reports easily

3. Easy to manage historical data in a secure manner

4. Centralized database helps in avoiding conflicts

The DISTRIBUTORS MANAGEMENT TOOL is a software application which avoids more manual hours that need to spend in record keeping and generating reports. This application keeps the data in a centralized way which is available to all the users simultaneously. It is very easy to manage historical data in database. No specific training

is required for the distributors to use this application. They can easily use the tool that decreases

Manual hours spending for normal things and hence increases the performance. It is very easy to record the information of online sales and purchases in the databases

- **Reduced Repair Costs:** By addressing small issues before they become bigger problems, organizations can help reduce repair costs over time. This can help save money and increase profitability.

- **Extended Vehicle Lifespan:** Regular maintenance can help extend the lifespan of vehicles, reducing the need for replacement and saving money in the long term.

- **Improved Resale Value:** Vehicles that have been well-maintained and have a documented maintenance history often have a higher resale value than those that have not. This can help organizations recoup some of the costs of their fleet of vehicles when it comes time to sell them.

- **Increased Efficiency**: Vehicles that are well-maintained are often more efficient, leading to lower fuel costs and increased productivity. This can help organizations save money and increase profits.

CHAPTER-4

SYSTEM REQUIREMENTS

4.1 FUNCTIONAL REQUIREMENTS

- **Admin**

In this application admin is the module, here admin can login directly with the application and after login successful admin can perform operations such as ViewAllUsers and addServices add view Services and logout.

- **Services**

In this application services is a module, here service should be added by the admin then only the services can access his home page after successful login he can perform some operations such as addCarDetails and logout

- **User**

In this application user is a module, here user should register with the application then only the user can access his home page after successful login he can perform some operations such as viewProfile, searchCarDetails, analysisGraph and logout.

4.2 NON-FUNCTIONAL REQUIREMENTS

4.2.1 HARDWARE REQUIREMENTS

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Floppy Drive : 1.44 Mb.
- Monitor : 15 VGA Colour.
- Mouse : Logitech.
- Ram : 1GB

4.2.2 **SOFTWARE REQUIREMENTS:**

- Operating system : Windows 10.
- Coding Language : JAVA
- Database : MySQL
- IDE : NETBEANS
- Platform : Windows

CHAPTER-5

SYSTEM STUDY

5.1 FEASIBILITY STUDY:

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

5.2 FEASIBILITY ANALYSIS

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

A feasibility analysis is undertaken to determine the possibility or probability of either improving the existing system or developing a completely new system.
It helps to obtain an overview of the problem and to get rough assessment of whether feasible solution exists.

There are three aspects in feasibility study portion of the preliminary investigation.

1) Operational feasibility
2) Technical feasibility
3) Economical feasibility

Operational feasibility :- AUTOMOBILE MANAGEMENT SYSTEM

It is a measure of how well a proposed system solves the problems, and takes advantages of the opportunities identified during scope definition and how it satisfies the requirements identified in the requirements analysis phase of system development.

Operational feasibility covers two aspects. One is the technical performance aspect and other is the acceptance within the organization. operational feasibility determines how the proposed system will fit the current operations and what, if any job restructuring and retraining may be needed to implement the system.

in the system operational feasibility checks, whether the user who is going to use the system is able to work with the software with which the system id coded and also the mind of the user going to use system. if the user does not understand or is able to work on the system further development is waste.

the system is easy to learn and it will require a vary short time to learn the operation of the system for a person having knowledge in accounting. so that system was operationally feasibl

after conducting the surveys in online and offline we get some responses on automobile management system.

•by getting the responses we analysis the problems.

no alt text provided for this image

he probles are that we have analysised are: some users don't know how to use internet, people preferance etc…

no alt text provided for this image

•these are the most common responses we have received.we want to reslove this problems in this sdp project.

reuirements specifications

•the first is we need the customesr support for the we want to introduce chat bot (like google asst).it will support/help to the customers.

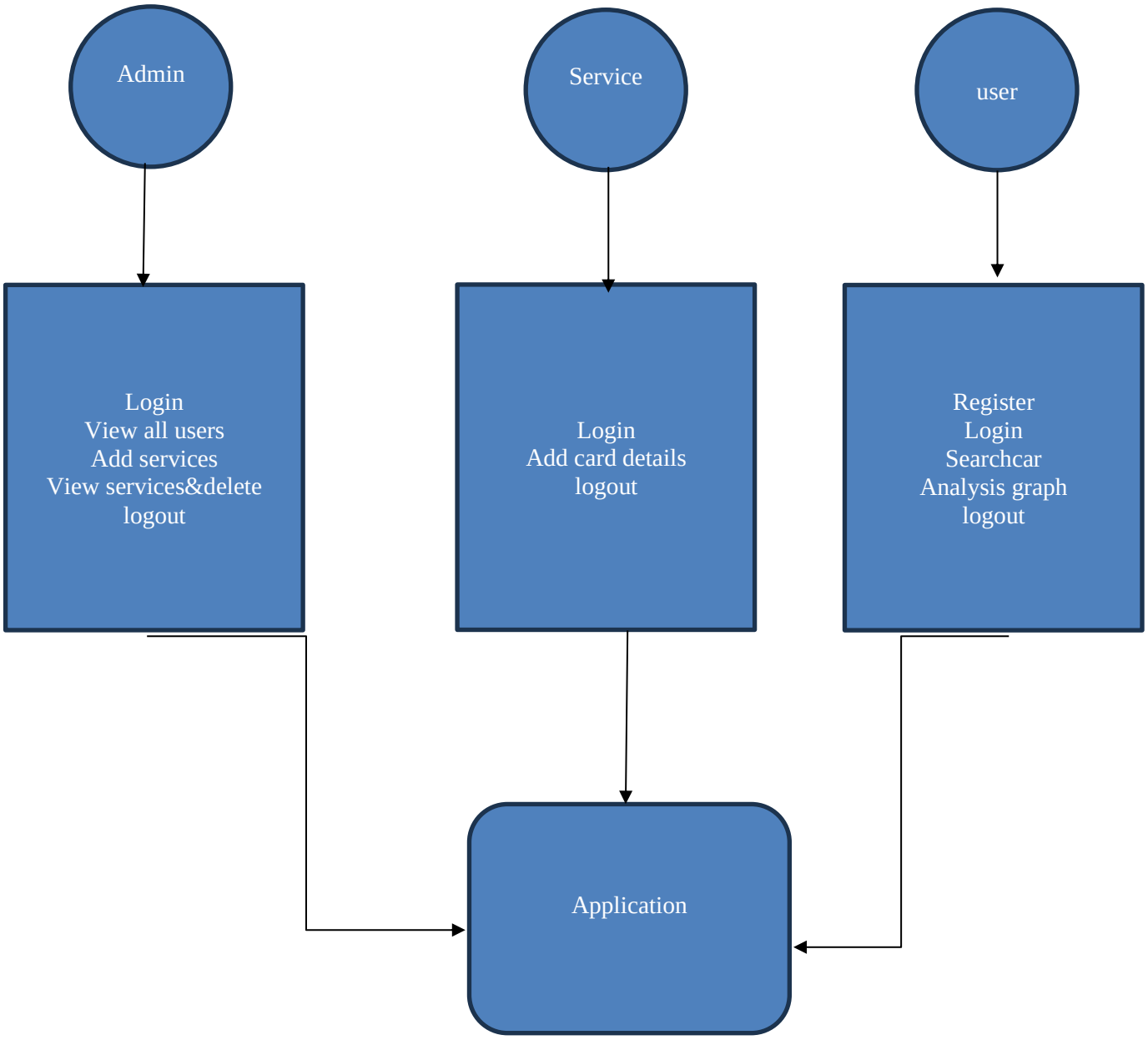
•for many of the websites the chat bot was introduced.we want to introduce in vehicle rental system also.

•the second thing is to clear the server problems.modules

CHAPTER-6

SYSTEM ARCHITECTURE

6.1 DATA FLOW DIAGRAMS



6.2 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components:

A Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.

2. Provide extendibility and specialization mechanisms to extend the core concepts

3. Be independent of particular programming languages and development process

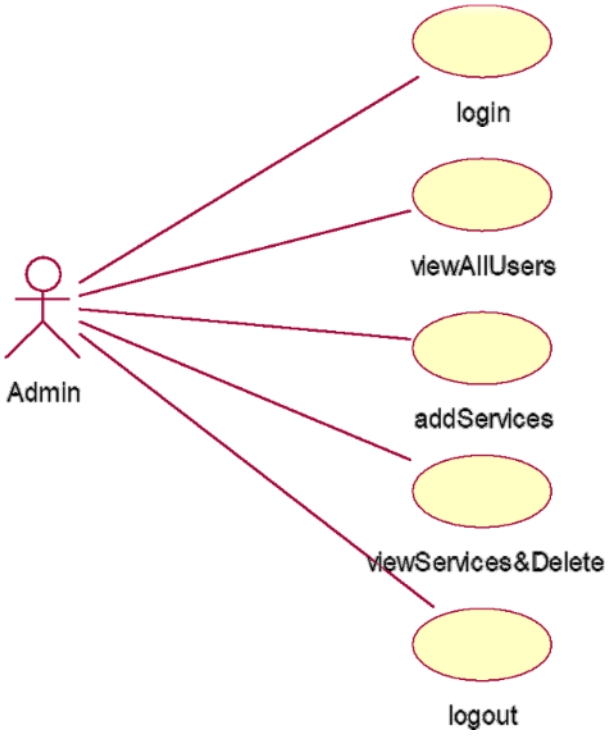
4. Provide a formal basis for understanding the modeling language.

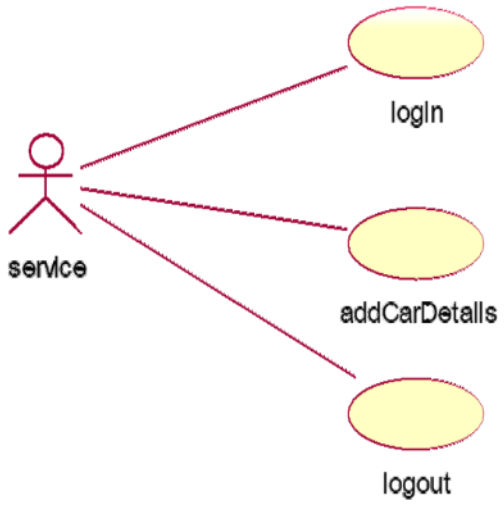
5. Encourage the growth of OO tools market.

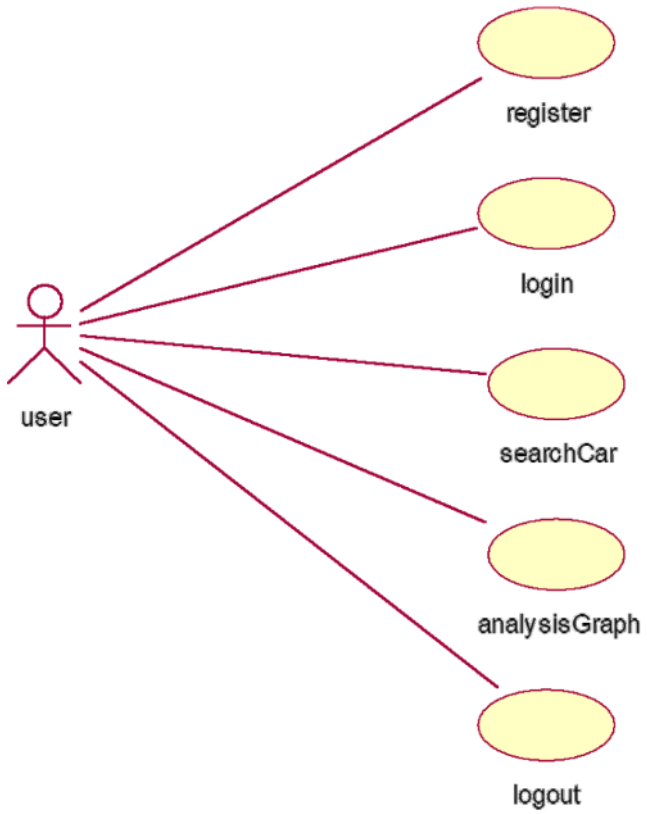
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.

6.2.1 USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases

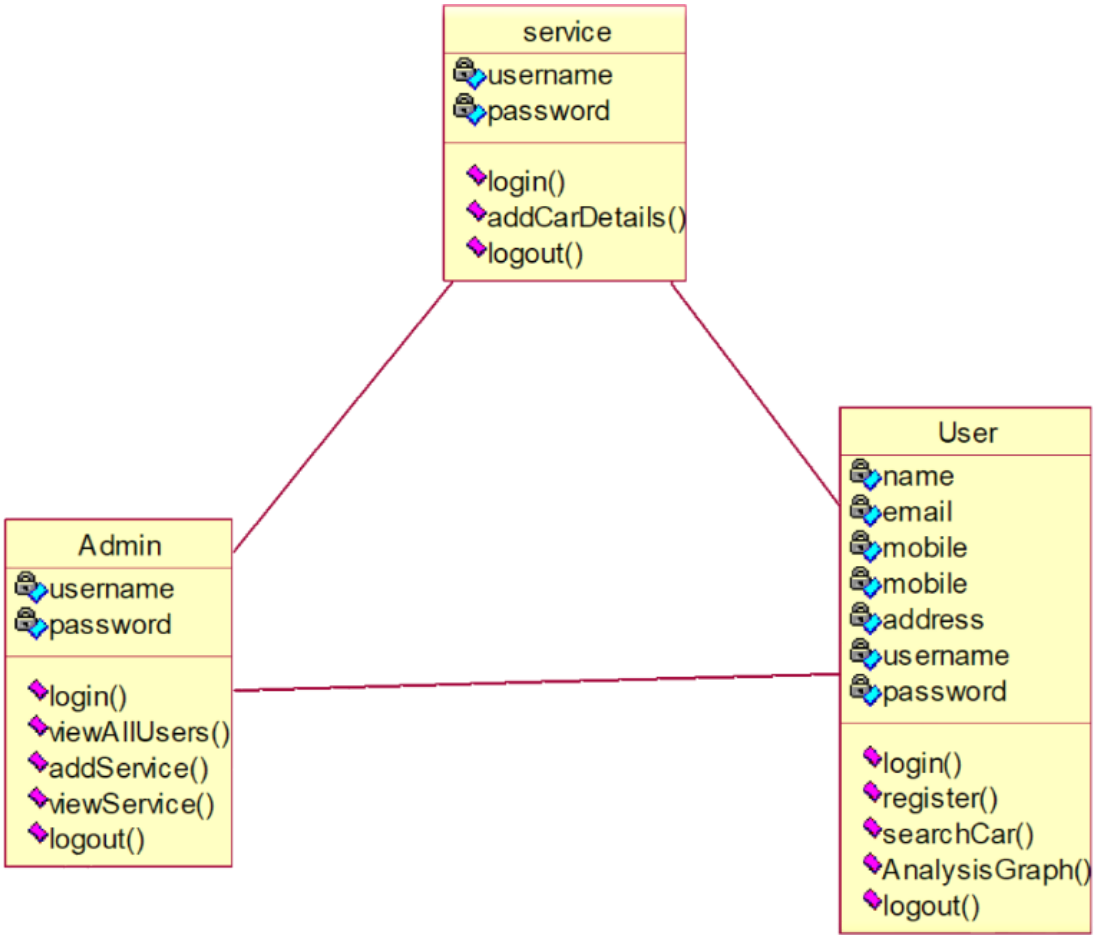



login
service
addCarDetails
logout


register
login
user
searchCar
analysisGraph
logout

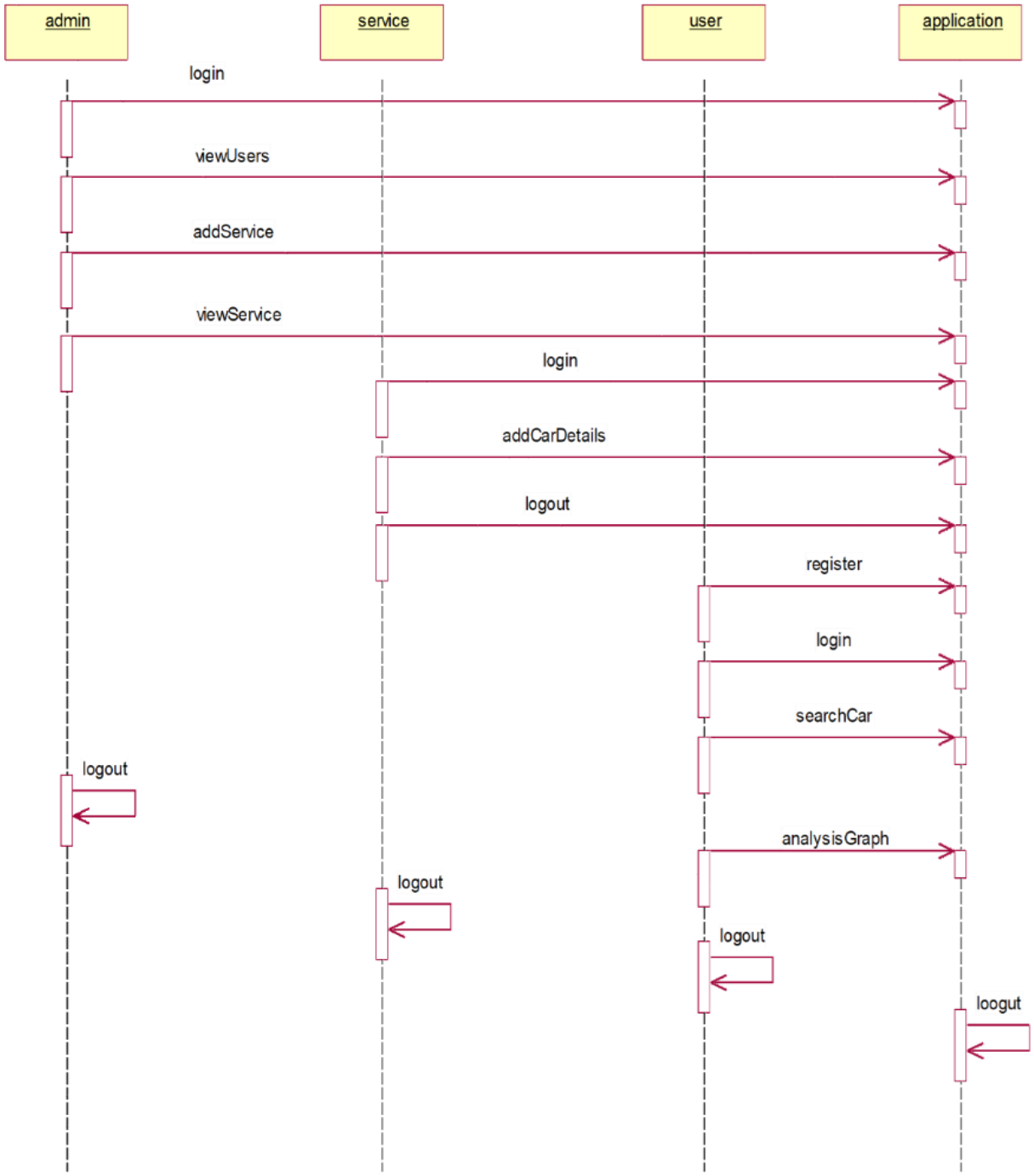
6.2.2 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.



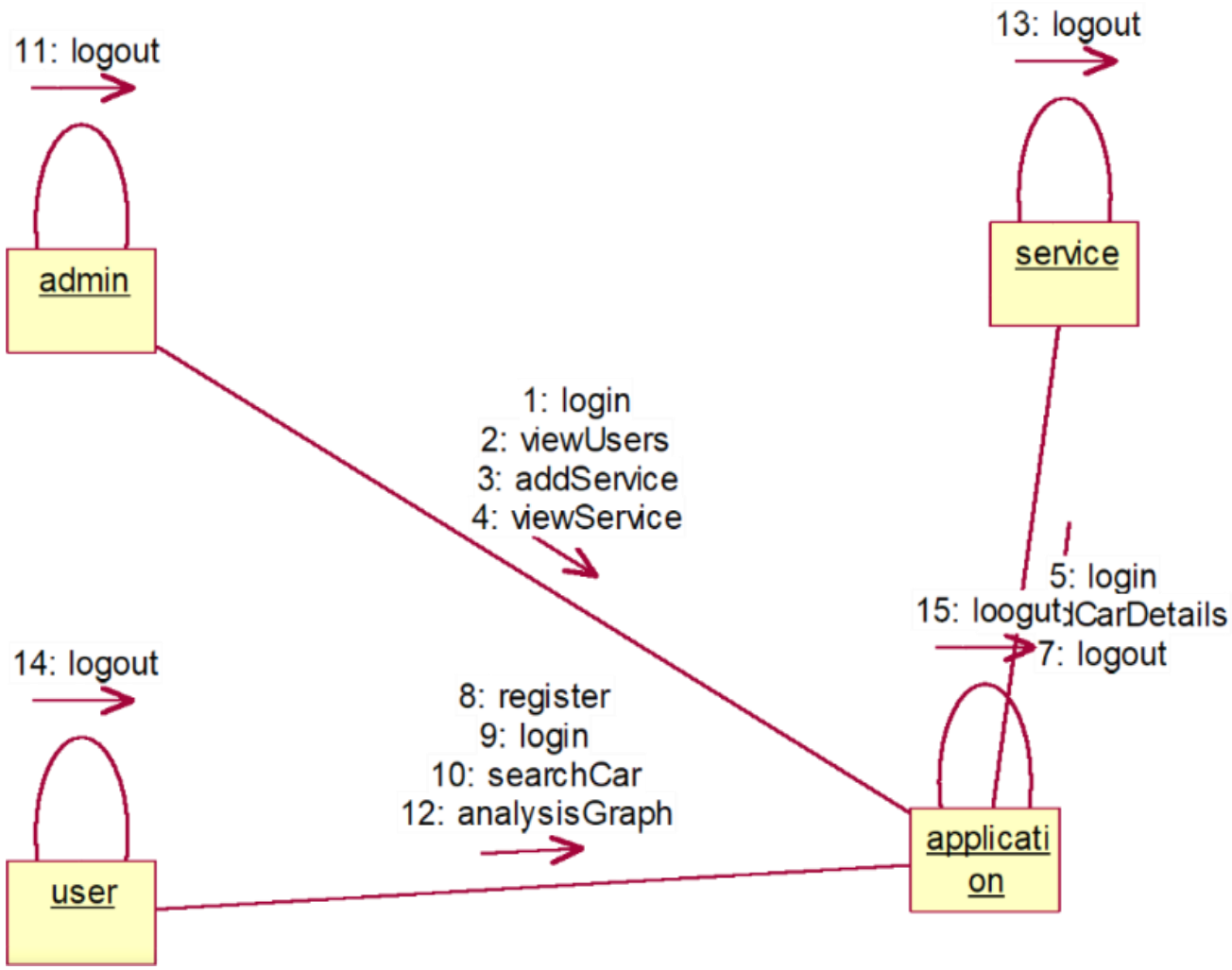
6.2.3 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.



6.2.4 COLLABORATION DIAGRAM

Collaboration is a working practice whereby individuals work together for a common purpose to achieve business benefit. Collaboration enables individuals to work together to achieve a defined and common business purpose.



6.2.5 ACTIVITY DIAGRAM

Activity diagram is another important diagram in UML to describe the dynamic aspects of the system.

Activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system.

The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent. Activity diagrams deal with all type of flow control by using different elements such as fork, join, etc

The basic purposes of activity diagrams is similar to other four diagrams. It captures the dynamic behavior of the system. Other four diagrams are used to show the message flow from one object to another but activity diagram is used to show message flow from one activity to another

Activity is a particular operation of the system. Activity diagrams are not only used for visualizing the dynamic nature of a system, but they are also used to construct the executable system by using forward and reverse engineering techniques. The only missing thing in the activity diagram is the message part.

It does not show any message flow from one activity to another. Activity diagram is sometimes considered as the flowchart. Although the diagrams look like a flowchart, they are not. It shows different flows such as parallel, branched, concurrent, and single.

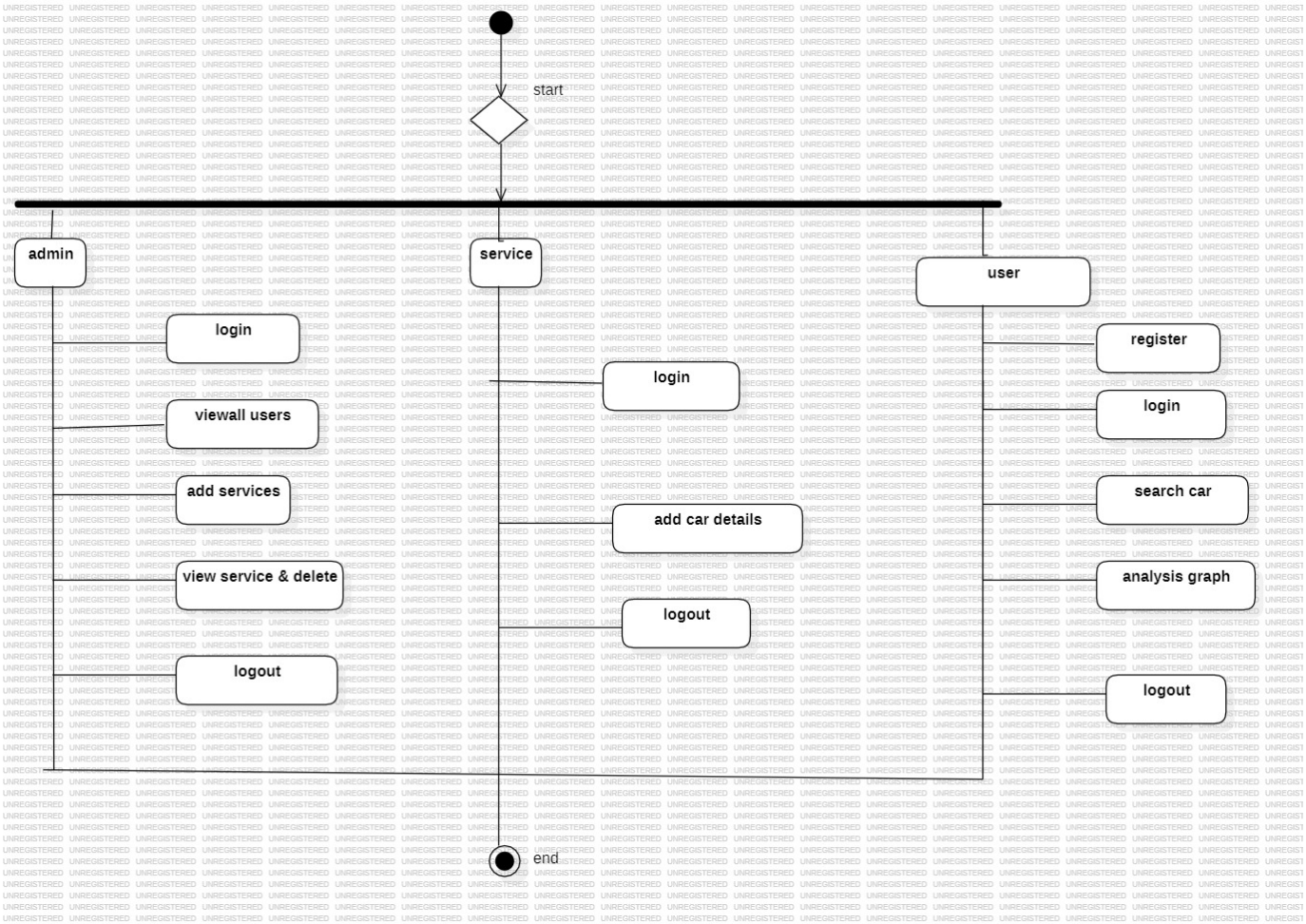
The basic usage of activity diagram is similar to other four UML diagrams. The specific usage is to model the control flow from one activity to another. This control flow does not include messages.

Activity diagram is suitable for modeling the activity flow of the system. An application can have multiple systems. Activity diagram also captures these systems and describes the flow from one system to another. This specific usage is not available in other diagrams. These systems can be database, external queues, or any other system.

We will now look into the practical applications of the activity diagram. From the above discussion, it is clear that an activity diagram is drawn from a very high level. So it gives high level view of a system. This high level view is mainly for business users or any other person who is not a technical person.

This diagram is used to model the activities which are nothing but business requirements.

The diagram has more impact on business understanding rather than on implementation details.

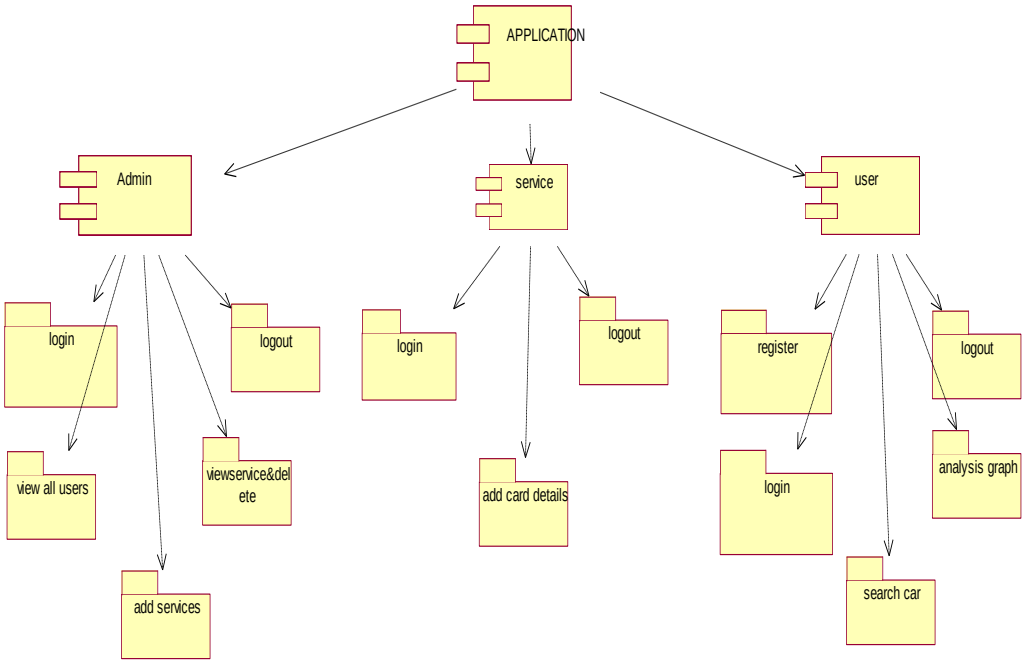


6.2.6 COMPONENT DIAGRAM

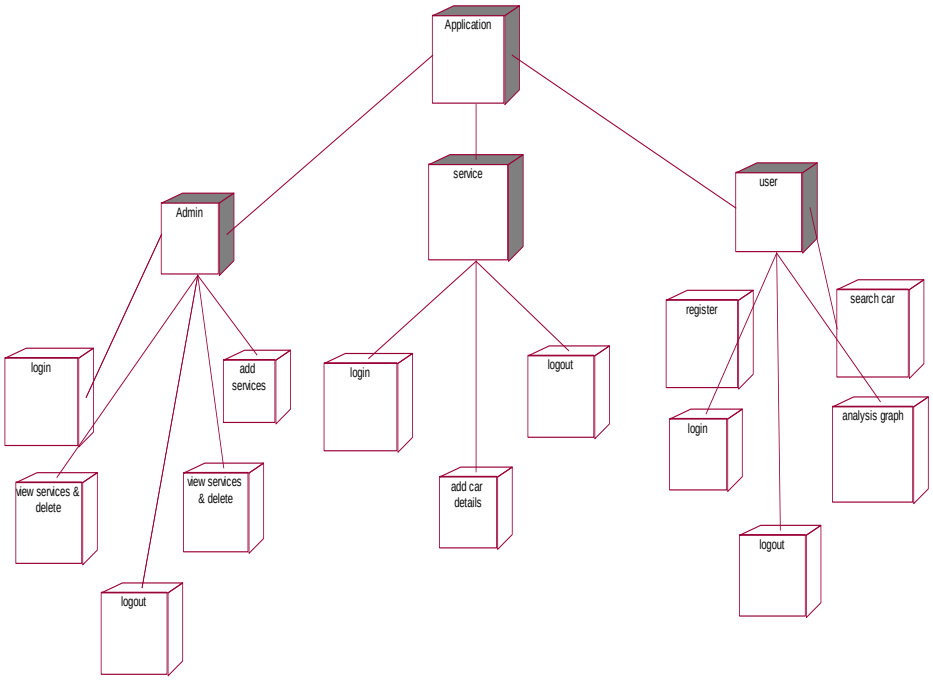
The data (account and inspection ID) flows into the component via the port on the right-hand side and is converted into a format the internal components can use. The interfaces on the right are known as required interfaces, which represents the services the component needed in order to carry out its duty.

The data then passes to and through several other components via various connections before it is output at the ports on the left. Those interfaces on the left are known as provided interface, which represents the services to deliver by the exhibiting component.

It is important to note that the internal components are surrounded by a large 'box' which can be the overall system itself (in which case there would not be a component symbol in the top right corner) or a subsystem or component of the overall system (in this case the 'box' is a component itself).



6.2.7 DEPLOYMENT DIAGRAM



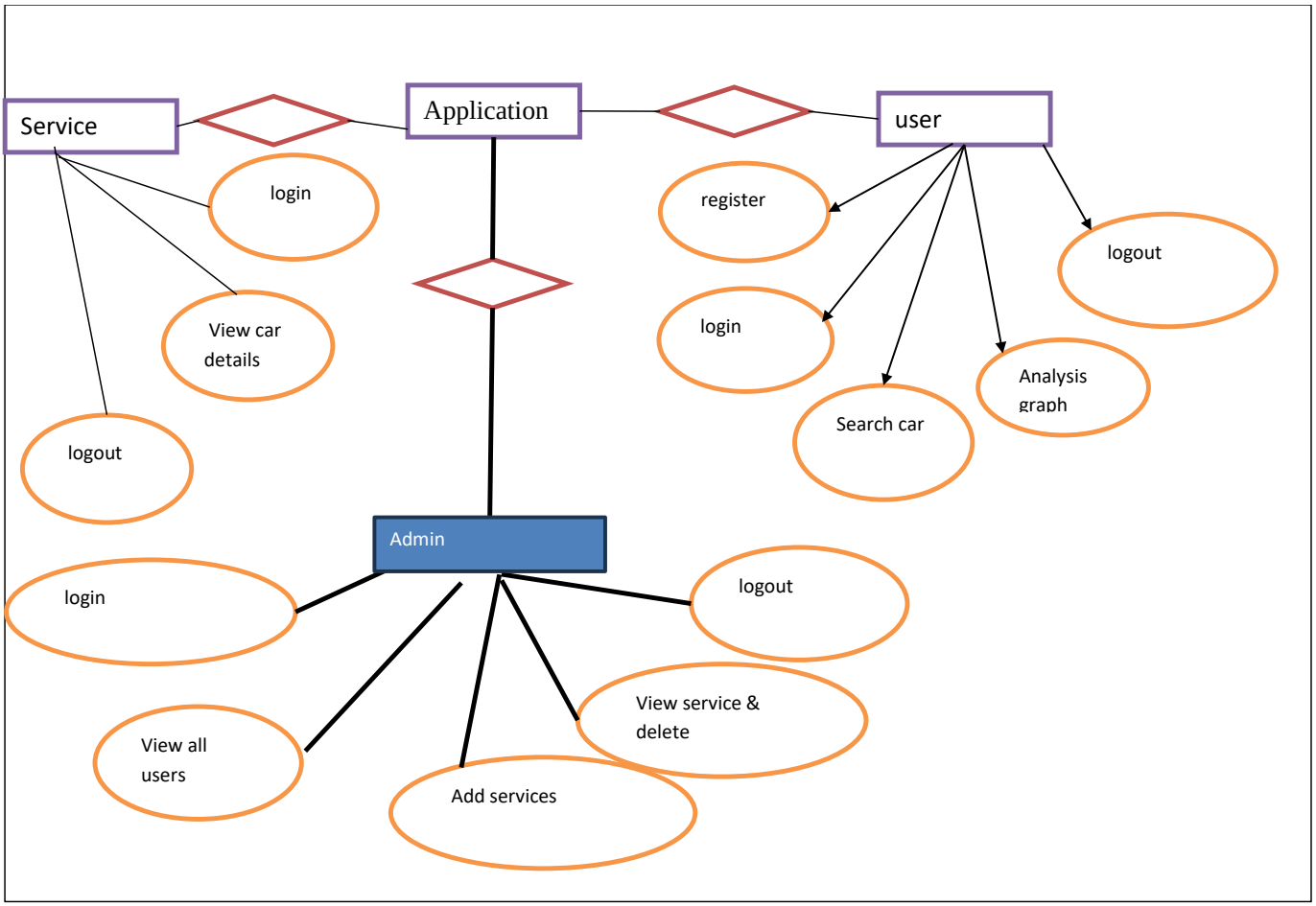
6.2.8 ER DIAGRAM

An Entity Relationship Diagram (ER Diagram) pictorially explains the relationship between entities to be stored in a database. Fundamentally, the ER Diagram is a structural design of the database. It acts as a framework created with specialized symbols for the purpose of defining the relationship between the database entities. ER diagram is created based on three principal components: entities, attributes, and relationships.

ER Diagram helps you conceptualize the database and lets you know which fields need to be embedded for a particular entity

- ER Diagram gives a better understanding of the information to be stored in a database
- It reduces complexity and allows database designers to build databases quickly
- It helps to describe elements using Entity-Relationship models
- It allows users to get a preview of the logical structure of the database

1. Attributes
2. Key Attribute
3. Composite Attribute
4. Multivalued Attribute
5. Derived Attribute
6. Relationships
7. One-to-One Relationships
8. One-to-Many Relationships
9. Many-to-One Relationships
10. Many-to-Many Relationships


Service
Application
user
login
View car details
logout
register
login
Search car
Analysis graph
logout
Admin
login
View all users
Add services
View service & delete
logout

6.2.9 DATA DICTIONARY :

A data dictionary contains metadata i.e., data about the database. The data dictionary is very important as it contains information such as what is in the database, who is allowed to access it, where is the database physically stored etc. The users of the database normally don't interact with the data dictionary, it is only handled by the database administrators.

A data dictionary Is a collection of descriptions of the data objects or items in a data model to which programmers and others can refer. Often, a data dictionary is a centralized metadata repository.

Data dictionaries sometimes play a role in data modeling, which creates a tangible diagram of object relationships that lists each object's name, assigned data values and defined relationships. The type of data, such as text, image or binary value, is described; possible predefined default values are listed; and a brief textual description is provided. This collection of information can be referenced through a data dictionary.

There are two types of data dictionaries: active and passive.

Active data dictionaries are created within the databases they describe and automatically reflect any updates or changes in their host databases. This avoids any discrepancies between the data dictionaries and their database structures.

Passive data dictionaries are created separately from the databases they describe to act as a repository for data information. Passive data dictionaries require additional work to stay in sync with the databases they describe. As such, database managers must handle passive directories with care to ensure there are no discrepancies.

The specific components of a data dictionary can vary, but they typically take the form of various types of metadata. Examples of these components include the following:

Data object listings, such as names and definitions;

Data element properties, such as data types, unique identifiers, sizes and indexes;

Entity relationship diagrams;

System-level diagrams;

Reference data;
Missing data and quality indicator codes; and

Business rules for validation of data quality and schema objects.

When planning a data dictionary build, it is important to consider all available data management resources, including associated databases and spreadsheets.

Most database management systems, as well as information systems created by computer-aided software engineering tools, contain integrated active data dictionaries. For example, the Performance Analyzer tool for Microsoft Access, which analyzes and documents databases, can create data dictionaries from data either based in or connected to a Microsoft Access implementation.
If it's not possible to automatically generate a machine-readable data dictionary, it's recommended to submit a data dictionary from a single source as a spreadsheet file. Spreadsheets can be made into data dictionaries within Microsoft Excel. There are also online templates that can come in handy when creating this type of data dictionar

Table name: automobile management system

Primary key: username

Column	Datatype	Constraints	Description
Username	Varchar(10)	NotNull	To enter username
Password	Int	NotNull	For

			security

PRIMARY KEY :

The SQL Primary Key is a column (or combination of columns) that uniquely identifies each record in a database table. The Primary Key also speeds up data access and is used to establish a relationship between tables.

It contains only a unique value.

It can not be null.

One table can have only one Primary Key.

A primary key length cannot be more than 900 bytes.

While creating a table using the CREATE TABLE statement, you can add the primary key constraint on a particular column of the table just by to specifying the name of the column along with the keyword “PRIMARY KEY”.

A column or columns is called primary key (PK) that uniquely identifies each row in the table.

If you want to create a primary key, you should define a PRIMARY KEY constraint when you create or modify a table.

When multiple columns are used as a primary key, it is known as composite primary key.

Primary key enforces the entity integrity of the table.

Primary key always has unique data.

A primary key length cannot be exceeded than 900 bytes.

A primary key cannot have null value.

can be no duplicate value for a primary key.

A table can contain only one primary key constrain.

Field name	Datatype	Constraints	Description
Cid	Text	Not Null	Customer id
Cname	Text	Not Null	Customer name
Caddress	Text	Not Null	Address of the customer
Cgender	Text	Not Null	Gender of the customer
Cphone	Number	Not Null	Phone number of the customer
Ccity	Text	Not Null	City of the customer
Cemail	Text	Not Null	Email I'd of the customer
Clandline	Text	Not Null	Landlines number of the customer

User

Field name	Datatype	Name
User_id	Varchar	Primary key
User_name	Varchar	Primary key
Password	Varchar	Primary key
Address	Varchar	Primary key

Admin

Field name	Datatype	Name
Admin_id	Varchar	Primary key
Admin_name	Varchar	Primary key

Password	Varchar	Primary key

Vehicle

Field name	Data type	Name
Vehicle _id	Varchar	Primary key
Vehicle _type	Varchar	Primary key
Owner _name	Varchar	Primary key

The concept of self-propelled vehicles dates back centuries, but the late 19th and early 20th centuries marked the period of rapid innovation in automobile technology. Karl Benz is credited with building the first practical automobile powered by an internal combustion engine in 1885. This innovation set the wheels in motion for the automotive industry, leading to the production of mass-market automobiles by Henry Ford's assembly line techniques in the early 20th century.

As technology advanced, so did the design and capabilities of automobiles. The early 20th century witnessed the introduction of various features, such as electric starters, more efficient engines, and safety features like seat belts and airbags. These developments paved the way for automobiles to become an integral part of daily life for people worldwide

Driving into the Future: Innovations in Automobile Technology

The automotive industry has continuously pushed the boundaries of innovation, addressing concerns about environmental sustainability, safety, and efficiency. One of the most

significant recent developments is the rise of electric vehicles (EVs). With advancements in battery technology, EVs are becoming more accessible, offering zero-emission alternatives to traditional gasoline-powered cars.

Moreover, autonomous driving technology is poised to revolutionize transportation once again. Companies are investing heavily in self-driving systems, which have the potential to reduce accidents, alleviate traffic congestion, and provide newfound mobility for people with disabilities or those unable to drive.

Impact on Society and Environment

The widespread adoption of automobiles has had a profound impact on societies globally. The automobile industry has generated millions of jobs, driving economic growth and development. Cities and towns have transformed to accommodate cars, shaping urban planning and influencing infrastructure projects.

However, this transformation has not been without challenges. Concerns about air pollution, greenhouse gas emissions, and traffic congestion have prompted the industry to seek cleaner and more efficient transportation solutions. The shift towards electric vehicles and the development of hybrid technology reflects a commitment to reducing the environmental footprint of automobiles.

Cultural and Lifestyle Implications

Automobiles have also shaped cultural trends and lifestyles. The idea of personal freedom, exploration, and the open road is deeply intertwined with the concept of driving. Road trips, car culture, and iconic vehicles have become symbols of adventure, individualism, and even status.

Furthermore, the concept of car ownership is evolving with the rise of shared mobility services like ride-hailing and car-sharing platforms. These services are changing the way people perceive transportation, emphasizing access over ownership and potentially reducing the number of vehicles on the road

CHAPTER-7

INPUT OUTPUT DESIGNS

7.1 INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

1. What data should be given as input?
2. How the data should be arranged or coded?
3. The dialog to guide the operating personnel in providing input.
4. Methods for preparing input validations and steps to follow when error occur.

INPUT OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

7.2 OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system. The output form of an information system should accomplish one or more of the following objectives.

1. Convey information about past activities, current status or projections of the

2. Future.

3. Signal important events, opportunities, problems, or warnings.

4. Trigger an action

CHAPTER-8

IMPLEMENTATION

8.1 MODULE

1. **ADMIN**
2. **SERVICE**
3. **USER**

8.1.1 MODULE DESCRIPTION

Admin

In this application admin is the module, here admin can login directly with the application and after login successful admin can perform operations such as ViewAllUsers and addServices add view Services and logout.

1. User Access Control: The admin module should include features to manage user access levels and permissions within the system. This ensures that only authorized personnel can view or modify sensitive information.

2. Vehicle Database Management: The admin module should allow for the efficient management of the vehicle database, including adding new vehicles, updating information, and removing outdated records.

3. Maintenance Scheduling: Admins should be able to schedule and track vehicle maintenance tasks through the system. This feature helps in ensuring that vehicles are serviced regularly and remain in optimal condition.

Services

In this application services is a module, here service should be added by the admin then only the services can access his home page after successful login he can perform some operations such as addCarDetails and logout

1. Service Scheduling: The service module should allow users to schedule maintenance and service appointments for their vehicles. It should provide reminders for upcoming services based on mileage or time intervals.

2. Service History Tracking: Users should be able to view the service history of their vehicles within the system. This includes details of past maintenance tasks, repairs, and any replacements done.

3. Cost Estimation: The service module can include a feature for estimating the cost of upcoming services or repairs based on the type of service needed and the vehicle's make and model.

User

In this application user is a module, here user should register with the application then only the user can access his home page after successful login he can perform some operations such as viewProfile, searchCarDetails, analysisGraph and logout.

1. User Profile Management: The user module allows users to create and manage their profiles within the system. Users can update personal information, contact details, and vehicle preferences.

2. Vehicle Registration: Users can register their vehicles in the system, providing details such as make, model, year, and registration information. This helps in tracking and managing individual vehicles.

3. Service Reminders: The user module can send service reminders to users based on their vehicle's maintenance schedule. This feature ensures that users stay up-to-date with their vehicle's service needs.

CHAPTER-9

SOFTWARE ENVIRONMENT

9.1 JAVA TECHNOLOGY

Java technology is both a programming language and a platform.

The Java Programming Language

The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

1. Simple
2. Architecture neutral
3. Object oriented
4. Portable
5. Distributed
6. High performance
7. Interpreted
8. Multithreaded
9. Robust
10. Dynamic
11. Secure

Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be a bit slower than native code. However, smart compilers, well-tuned interpreters, and just-in-time byte

code compilers can bring performance close to that of native code without threatening portability.

What Can Java Technology Do?

The most common types of programs written in the Java programming language are applets and applications. If you've surfed the Web, you're probably already familiar with applets. An applet is a program that adheres to certain conventions that allow it to run within a Java-enabled browser.

However, the Java programming language is not just for writing cute, entertaining applets for the Web. The general-purpose, high-level Java programming language is also a powerful software platform. Using the generous API, you can write many types of programs.

An application is a standalone program that runs directly on the Java platform. A special kind of application known as a server serves and supports clients on a network. Examples of servers are Web servers, proxy servers, mail servers, and print servers. Another specialized program is a servlet. A servlet can almost be thought of as an applet that runs on the server side. Java Servlets are a popular choice for building interactive

Auto Machine is a solution to automate the activities of an Automobile Dealers. Developed with a lot of research in the automobile industry, Auto Machine provides a simple but robust solution to manage your organization with ease. Sales and Inventory being the key areas of focus for automobile dealers, utmost care has been taken to include even the minutest action and information. With ease of navigation and one screen access to all functionalities, Auto Machine presents an environment where in the dealers can concentrate on Sales and Service with the assurance that all the data is extremely secure and available immediately when needed. With our centralized system, branch management, dealer management and stock transfers has become simple and fast.Web applications, replacing the use of CGI scripts. Servlets are similar to applets in that they are runtime extensions of applications. Instead of working in browsers, though, servlets

run within Java Web servers, configuring or tailoring the server.

How does the API support all these kinds of programs? It does so with packages of software components that provides a wide range of functionality. Every full implementation of the Java platform gives you the following features:

1. The essentials: Objects, strings, threads, numbers, input and output, data structures, system properties, date and time, and so on.

2. Applets: The set of conventions used by applets.

3. Networking: URLs, TCP (Transmission Control Protocol), UDP (User Data gram Protocol) sockets, and IP (Internet Protocol) addresses.

4. Internationalization: Help for writing programs that can be localized for users worldwide. Programs can automatically adapt to specific locales and be displayed in the appropriate language.

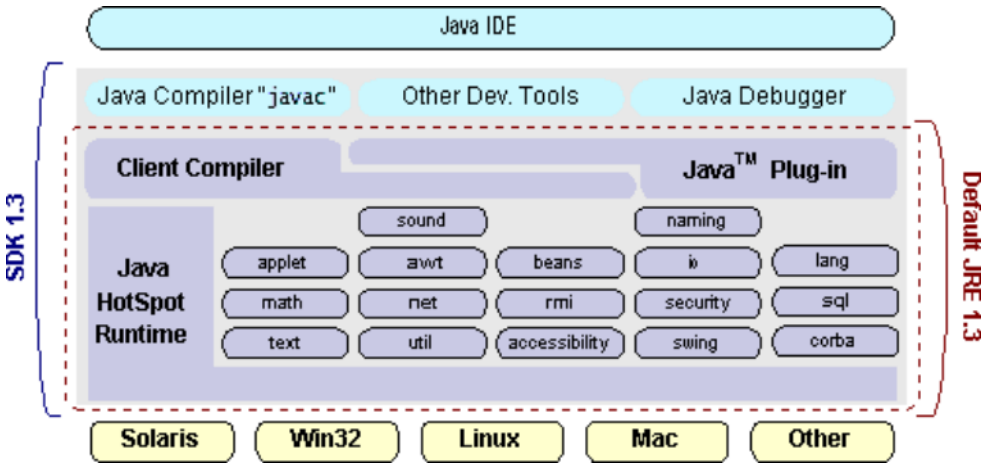
5. Security: Both low level and high level, including electronic signatures, public and private key management, access control, and certificates.

6. Software components: Known as JavaBeansTM, can plug into existing component architectures.

7. Object serialization: Allows lightweight persistence and communication via Remote Method Invocation (RMI).

8. Java Database Connectivity (JDBCTM): Provides uniform access to a wide range of relational databases.

a. The Java platform also has APIs for 2D and 3D graphics, accessibility, servers, collaboration, telephony, speech, animation, and more. The following figure depicts what is included in the Java 2 SDK.



How Will Java Technology Change My Life?

We can't promise you fame, fortune, or even a job if you learn the Java programming language. Still, it is likely to make your programs better and requires less effort than other languages. We believe that Java technology will help you do the following:

- Get started quickly: Although the Java programming language is a powerful object-oriented language, it's easy to learn, especially for programmers already familiar with C or C++.

- Write less code: Comparisons of program metrics (class counts, method counts, and so on) suggest that a program written in the Java programming language can be four times smaller than the same program in C++.

- Write better code: The Java programming language encourages good coding practices, and its garbage collection helps you avoid memory leaks. Its object orientation, its JavaBeans component architecture, and its wide-ranging, easily extendible API let you reuse other people's tested code and introduce fewer bugs.

- Develop programs more quickly: Your development time may be as much as twice as fast versus writing the same program in C++. Why? You write fewer lines of code and it is a simpler programming language than C++.

Through the ODBC Administrator in Control Panel, you can specify the particular database that is associated with a data source that an ODBC application program is written to use. Think of an ODBC data source as a door with a name on it. Each door will lead you to a particular database. For example, the data source named Sales Figures might be a SQL Server database, whereas the Accounts Payable data source could refer to an Access database. The physical database referred to by a data source can reside anywhere on the LAN.

The ODBC system files are not Installed on your system by Windows 95. Rather, they are installed when you setup a separate database application, such as SQL Server Client or Visual Basic 4.0. When the ODBC icon is installed in Control Panel, it uses a file called ODBCINST.DLL. It is also possible to administer your ODBC data sources through a stand- alone program called ODBCADM.EXE. There is a 16-bit and a 32-bit version of this program and each maintains a separate list of ODBC data sources.

From a programming perspective, the beauty of ODBC is that the application can be written to use the same set of function calls to interface with any data source, regardless of the database vendor. The source code of the application doesn't change whether it talks to Oracle or SQL Server. We only mention these two as an example. There are ODBC drivers available for several dozen popular database systems. Even Excel spreadsheets and plain text files can be turned into data sources. The operating system uses the Registry information written by

ODBC Administrator to determine which low-level ODBC drivers are needed to talk to the data source (such as the interface to Oracle or SQL Server). The loading of the ODBC drivers is transparent to the ODBC application program. In a client/server environment, the ODBC API even handles many of the network issues for the application programmer

The advantages of this scheme are so numerous that you are probably thinking there must be some catch. The only disadvantage of ODBC is that it isn't as efficient as talking directly to the native database interface. ODBC has had many detractors make the charge

that it is too slow. Microsoft has always claimed that the critical factor in performance is the quality of the driver software that is used. In our humble opinion, this is true. The availability of good ODBC drivers has improved a great deal recently. And anyway, the criticism about performance is somewhat analogous to those who said that compilers would never match the speed of pure assembly language. Maybe not, but the compiler (or ODBC) gives you the opportunity to write cleaner programs, which means you finish sooner. Meanwhile, computers

9.2 JDBC

In an effort to set an independent database standard API for Java; Sun Microsystems developed Java Database Connectivity, or JDBC. JDBC offers a generic SQL database access mechanism that provides a consistent interface to a variety of RDBMSs. This consistent interface is achieved through the use of “plug-in” database connectivity modules, or drivers. If a database vendor wishes to have JDBC support, he or she must provide the driver for each platform that the database and Java run on.

To gain a wider acceptance of JDBC, Sun based JDBC’s framework on ODBC. As you discovered earlier in this chapter, ODBC has widespread support on a variety of platforms. Basing JDBC on ODBC will allow vendors to bring JDBC drivers to market much faster than developing a completely new connectivity solution.

JDBC was announced in March of 1996. It was released for a 90 day public review that ended June 8, 1996. Because of user input, the final JDBC v1.0 specification was released soon after.

Few software packages are designed without goals in mind. JDBC is one that, because of its many goals, drove the development of the API. These goals, in conjunction with early reviewer feedback, have finalized the JDBC class library into a solid framework for building database applications in Java.

The goals that were set for JDBC are important. They will give you some insight as to why certain classes and functionalities behave the way they do. The eight design goals for JDBC are as follows:

1. SQL Level API

The designers felt that their main goal was to define a SQL interface for Java. Although not the lowest database interface level possible, it is at a low enough level for higher-level tools and APIs to be created. Conversely, it is at a high enough level for application programmers to use it confidently. Attaining this goal allows for future tool vendors to "generate" JDBC code and to hide many of JDBC's complexities from the end user.

2. SQL Conformance

SQL syntax varies as you move from database vendor to database vendor. In an effort to support a wide variety of vendors, JDBC will allow any query statement to be passed through it to the underlying database driver. This allows the connectivity module to handle non-standard functionality in a manner that is suitable for its users.

3. JDBC must be implemental on top of common

4. **database interfaces The JDBC SQL API** must "sit" on top of other common SQL level APIs. This goal allows JDBC to use existing ODBC level drivers by the use of a software interface.

This interface would translate JDBC calls to ODBC and vice versa.

5. Provide a Java interface that is consistent with the rest of the Java system

Because of Java's acceptance in the user community thus far, the designers feel that they should not stray from the current design of the core Java system.

6. Keep it simple

This goal probably appears in all software design goal listings. JDBC is no exception. Sun felt that the design of JDBC should be very simple, allowing for only one method of completing a task per mechanism. Allowing duplicate functionality only serves to confuse the users of the API.

7. Use strong, static typing wherever possible

Strong typing allows for more error checking to be done at compile time; also, less error appear at runtime.

8. Keep the common cases simple

Because more often than not, the usual SQL calls used by the programmer are simple SELECT's, INSERT's, DELETE's and UPDATE's, these queries should be simple to perform with JDBC. However, more complex SQL statements should also be possible.

Finally we decided to proceed the implementation using Java Networking. And for dynamically updating the cache table we go for MS Access database.

Java ha two things: a programming language and a platform.

Java is a high-level programming language that is all of the following

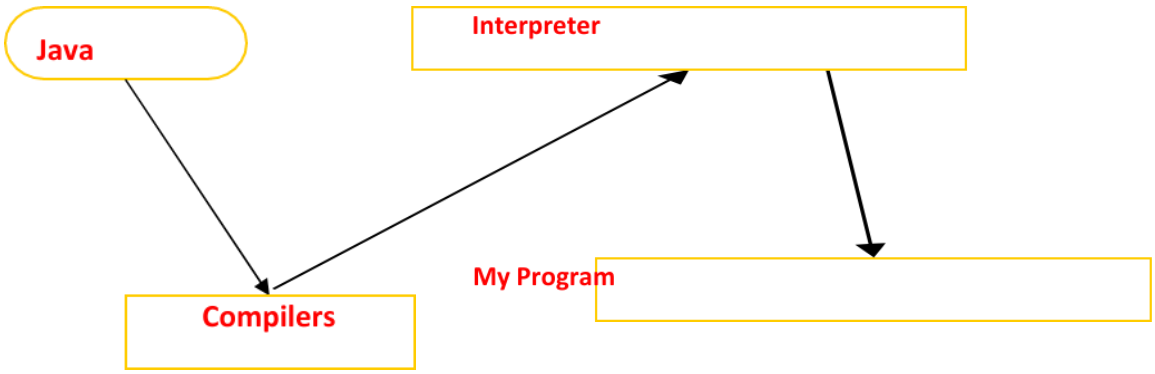
Simple Architecture-neutral Object-oriented Portable

Distributed High-performance

Interpreted

multithreaded Robust. Dynamic

Secure



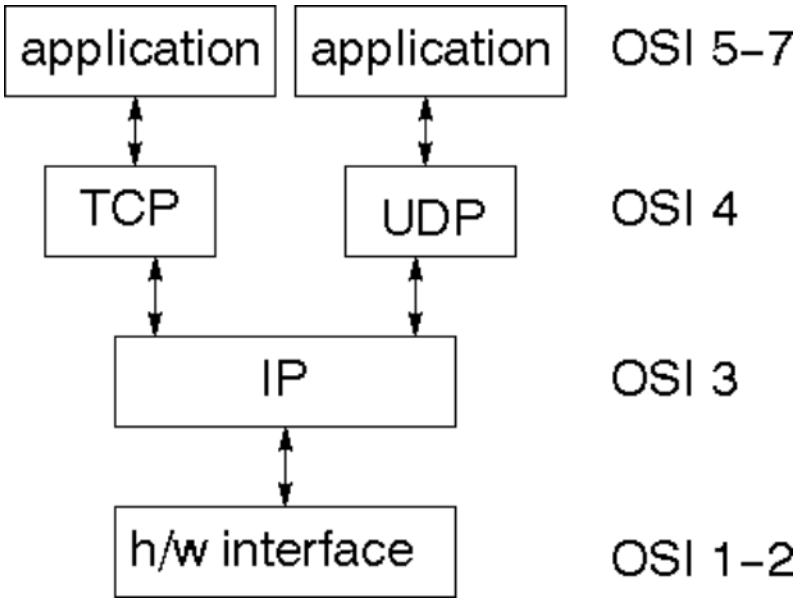
You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a Java development tool or a Web browser that can run Java applets, is an implementation of the Java VM. The Java VM can also be implemented in hardware.

Java byte codes help make "write once, run anywhere" possible. You can compile your Java program into byte codes on my platform that has a Java compiler. The byte codes can then be run any implementation of the Java VM. For example, the same Java program can run Windows NT, Solaris, and Macintosh.

Networking

TCP/IP stack

The TCP/IP stack is shorter than the OSI one:



TCP is a connection-oriented protocol; UDP (User Datagram Protocol) is a connectionless protocol.

IP datagram's

The IP layer provides a connectionless and unreliable delivery system. It considers each datagram independently of the others. Any association between datagram must be supplied by the higher layers. The IP layer supplies a checksum that includes its own header. The header includes the source and destination addresses. The IP layer handles routing through an Internet. It is also responsible for breaking up large datagram into smaller ones for transmission and reassembling them at the other end.++++++++++++++++++++++++++++++++++

UDP

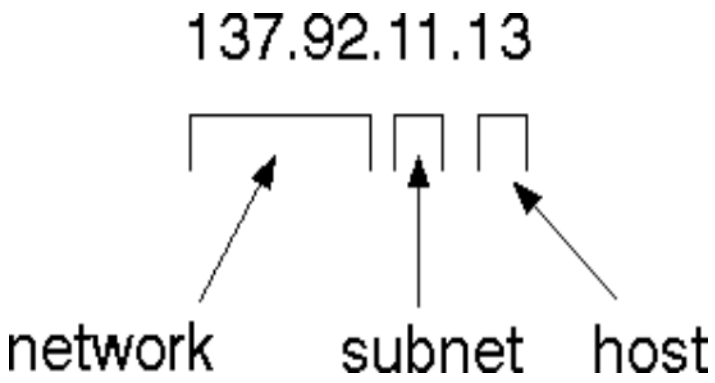
UDP is also connectionless and unreliable. What it adds to IP is a checksum for the contents of the datagram and port numbers. These are used to give a client/server model – see later.

TCP

TCP supplies logic to give a reliable connection-oriented protocol above IP. It provides a virtual circuit that two processes can use to communicate.

Internet addresses

In order to use a service, you must be able to find it. The Internet uses an address scheme for machines so that they can be located. The address is a 32 bit integer which gives the IP address. This encodes a network ID and more addressing. The network ID falls into various classes according to the size of the network address.



The 32 bit address is usually written as 4 integers separated by dots

Port addresses

A service exists on a host, and is identified by its port. This is a 16 bit number. To send a message to a server, you send it to the port for that service of the host that it is running on. This is not location transparency! Certain of these ports are “well known”.

Sockets

A socket is a data structure maintained by the system to handle network connections. A socket is created using the call socket. It returns an integer that is like a file descriptor. In fact, under Windows, this handle can be used with Read File and Write File functions.

#include <sys/types.h> #include <sys/socket.h>

Int socket(int family, int type, int protocol);

Here “family” will be AF_INET for IP communications, protocol will be zero, and type will depend on whether TCP or UDP is used. Two processes wishing to communicate over a network create a socket each. These are similar to two ends of a pipe – but the actual pipe does not yet exist.

JFree Chart

JFreeChart is a free 100% Java chart library that makes it easy for developers to display professional

quality charts in their applications. JFreeChart's extensive feature set includes:

A consistent and well-documented API, supporting a wide range of chart types; A flexible design that is easy to extend, and targets both server-side and client.

Side applications;

Support for many output types, including Swing components, image files (including PNG and JPEG), and vector graphics file formats (including PDF, EPS and SVG);

JFreeChart is "open source" or, more specifically, free software. It is distributed under the terms of the GNU Lesser General Public Licence (LGPL), which permits use in proprietary applications.

1. Map Visualizations

Charts showing values that relate to geographical areas. Some examples include: (a) population density in each state of the United States, (b) income per capita for each country in Europe, (c) life expectancy in each country of the world. The tasks in this project include:

Sourcing freely redistributable vector outlines for the countries of the world, states/provinces in particular countries (USA in particular, but also other areas);

Creating an appropriate dataset interface (plus default implementation), a rendered, and integrating this with the existing XYPlot class in JFreeChart;

Testing, documenting, testing some more, documenting some more.

2. Time Series Chart Interactivity

Implement a new (to JFreeChart) feature for interactive time series charts --- to display a separate control that shows a small version of ALL the time series data, with a sliding "view" rectangle that allows you to select the subset of the time series data to display in the main chart.

3. Dashboards

There is currently a lot of interest in dashboard displays. Create a flexible dashboard mechanism that supports a subset of JFreeChart chart types (dials, pies, thermometers, bars, and lines/time series) that can be delivered easily via both Java Web Start and an applet.

4. Property Editors

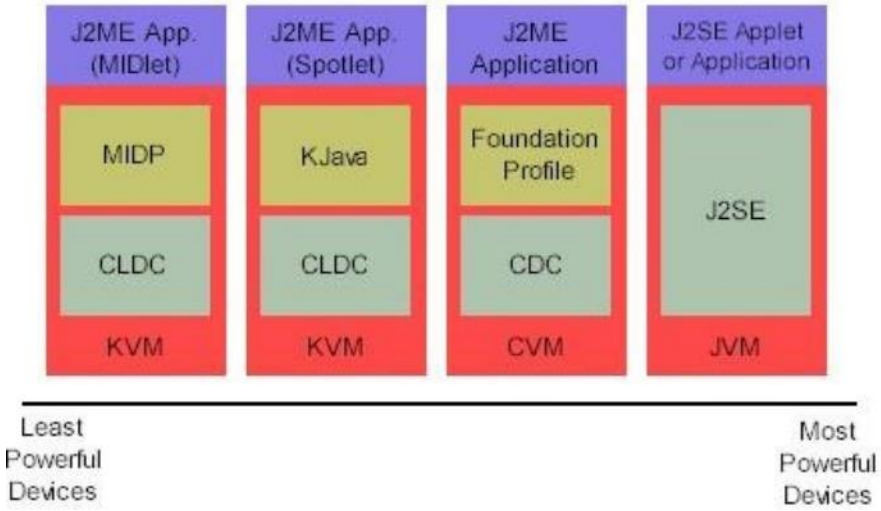
The property editor mechanism in JFreeChart only handles a small subset of the properties that can be set for charts. Extend (or reimplement) this mechanism to provide greater end-user control over the appearance of the charts.

J2ME (Java 2 Micro edition):-

Sun Microsystems defines J2ME as “a highly optimized Java run-time environment targeting a wide range of consumer products, including pagers, cellular phones, screen-phones, digital set-top boxes and car navigation systems.” Announced in June 1999 at the JavaOne Developer Conference, J2ME brings the cross-platform functionality of the Java language to smaller devices, allowing mobile wireless devices to share applications. With J2ME, Sun has adapted the Java platform for consumer products that incorporate or are based on small computing devices.

General J2ME architecture

As we mentioned earlier in this tutorial, a profile defines the type of device supported. The Mobile Information Device Profile (MIDP), for example, defines classes for cellular



phones. It adds domain-specific classes to the J2ME configuration to define uses for similar devices. Two profiles have been defined for J2ME and are built upon CLDC: KJava and MIDP. Both KJava and MIDP are associated with CLDC and smaller devices. Profiles are built on top of configurations. Because profiles are specific to the size of the device (amount of memory) on which an application runs, certain profiles are associated with certain configurations.

A skeleton profile upon which you can create your own profile, the Foundation Profile, is available for CDC.

Profile 1: KJava

KJava is Sun's proprietary profile and contains the KJava API. The KJava profile is built on top of the CLDC configuration. The KJava virtual machine, KVM, accepts the same byte codes and class file format as the classic J2SE virtual machine. KJava contains a Sun-specific API that runs on the Palm OS. The KJava API has a great deal in common with the J2SE Abstract Windowing Toolkit (AWT). However, because it is not a standard J2ME package, its main

Package is com.sun.kjava. We'll learn more about the KJava API later in this tutorial when we develop some sample applications.

Profile 2: MIDP

MIDP is geared toward mobile devices such as cellular phones and pagers. The MIDP, like KJava, is built upon CLDC and provides a standard run-time environment that allows new applications and services to be deployed dynamically on end user devices. MIDP is a common, industry-standard profile for mobile devices that is not dependent on a specific vendor. It is a complete and supported foundation for mobile application

Development. MIDP contains the following packages, the first three of which are core CLDC packages, plus three MIDP-specific packages.

1. Java.lang
2. Java.io
3. Java.util
4. Javax.microedition.io
5. Javax.microedition.lcdui
6. Javax.microedition.midlet
7. Javax.microedition.rms

The first key activity in the risk management process is Risk Identification. While in some publications ‘risk assessment” is used as an umbrella term that includes the primary activities of both risk identification and risk analysis this guide addresses these two critical risk management activities separately in Sections Error! Reference source not found. And Error! Reference source not found., respectively.

Purpose

The intent of risk identification is to answer the question “What can go wrong?” by:

Looking at current and proposed staffing, process, design, supplier, operational employment, resources, dependencies, etc.,Monitoring test results especially test failures

(readiness results and readiness problems for the sustainment phase),Reviewing potential shortfalls against expectations, and

• Analyzing negative trends.

Risk identification is the activity that examines each element of the program to identify associated root causes, begin their documentation, and set the stage for their successful management. Risk identification begins as early as possible in successful programs and continues throughout the program with regular reviews and analyses of Technical Performance Measurements (TPMS), schedule, resource data, life-cycle cost information, EVM data/trends, progress against critical path, technical baseline maturity, safety, operational readiness, and other program information available to program IPT members.

Tasks

Risk can be associated with all aspects of a program, e.g., operational needs, attributes, constraints, performance parameters including Key Performance Parameters (KPPs), threats, technology, design processes, or WBS elements. Consequently it is important to recognize that risk identification is the responsibility of every member of the IPT, not just the PM or systems engineer.

Examination of a program is accomplished through decomposition into relevant elements or areas. Decomposition may be oriented to requirements, processes, functional areas, technical baselines, or acquisition phases. Another method is to create a WBS as early as possible in a program for a product-oriented decomposition, which is particularly useful in identifying product and some process oriented risks. Other means, such as a process-oriented framework, would be required to sufficiently illuminate process-based root causes, which could be tracked via the WBS structure to view impacts to schedule, resource loading, etc.

To identify risks and their root causes, IPTs should break down program elements to a level where subject matter experts (SMEs) can perform valid identification by WBS or IMS line item number.

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To identify risks and their root causes, IPTs should break down program elements to a level where subject matter experts (SMEs) can perform valid identification by WBS or IMS line item number.

The information necessary to do this varies according to the life-cycle phase of the program. A program risk assessment checklist is available via the DAU (key words: “technical reviews;” course number CLE003).

During decomposition, risks can be identified based on prior experience, brainstorming, lessons learned from similar programs, and guidance contained in the program office RMP (see Section Error! Reference source not found.). A structured approach describes each WBS element in terms of sources or areas of risk, serves as the basis for identifying the first three levels of the program WBS, and developing the contract WBS. The examination

of each element and process against each risk area is an exploratory exercise to identify the critical root causes. The investigation may show that risks are inter-related.

WBS product and process elements and industrial engineering, manufacturing and repair processes are often sources of significant root causes. Risks are determined by examining each WBS element and process in terms of causes, sources, or areas of risk. When EVM is applied on a contract it can help identify WBS program elements that are experiencing issues. This information can be used to help prioritize WBS elements that may contain unidentified risks.

Risk should be continually reassessed (iterative) such as in integrated change control activity, when working with resources, when dealing with issues.

. Information gathering techniques

- Brainstorming

- Delphi technique: Expert participate anonymously; facilitator use questionnaire; consensus may be reached in a few rounds; Help reduce bias in the data and prevent influence each others.

Interviewing: interviewing experts, stakeholders, experienced PM

Root cause analysis: Reorganizing the identified risk by their root cause may help identify more risks

Checklist analysis: checklist developed based on accumulated historical information from previous similar project

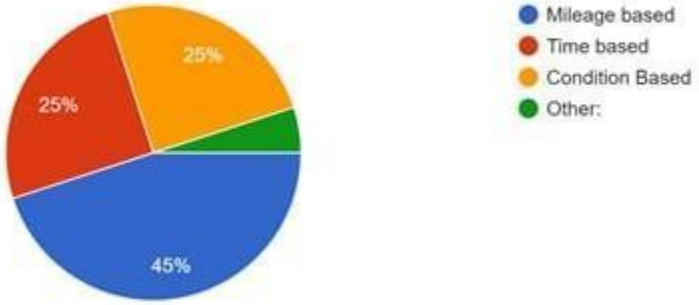
. Assumption analysis: identify risk from inaccuracy, instability, inconsistency, incompleteness.

- SWOT analysis – Strengths, Weaknesses, Opportunities, Threats

Would you like your automobile management system to integrate with other smart devices in your home?

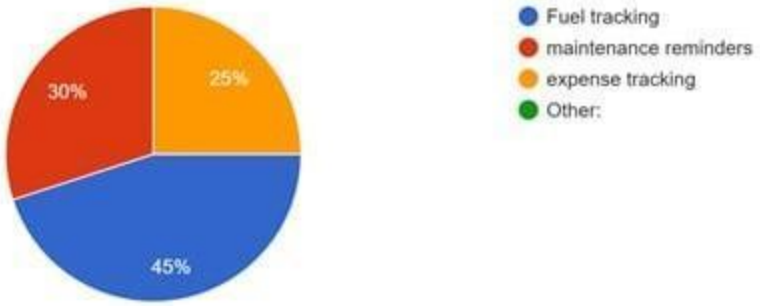
How often would you like to receive maintenance reminders for your automobile?

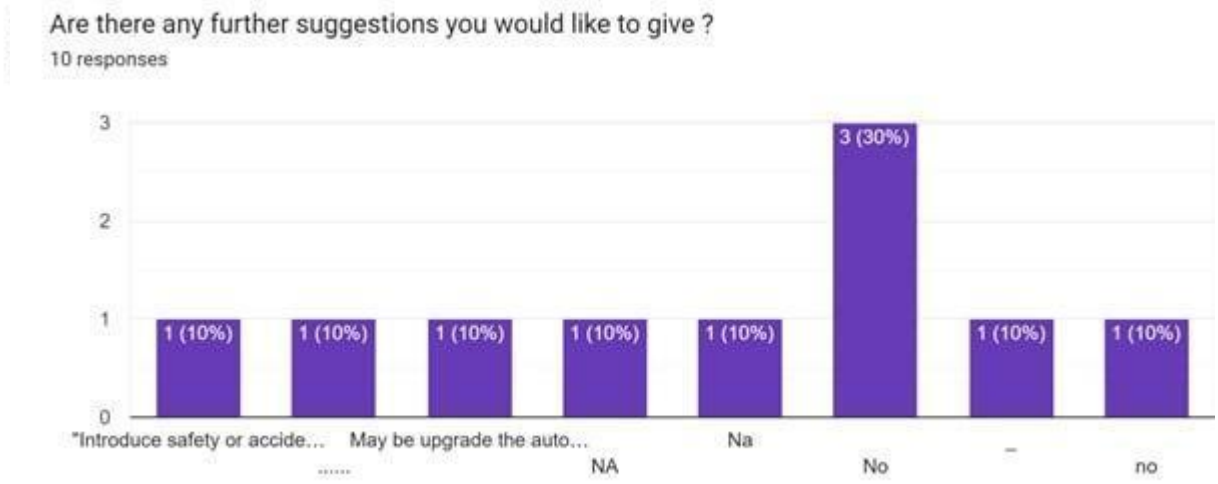
20 responses

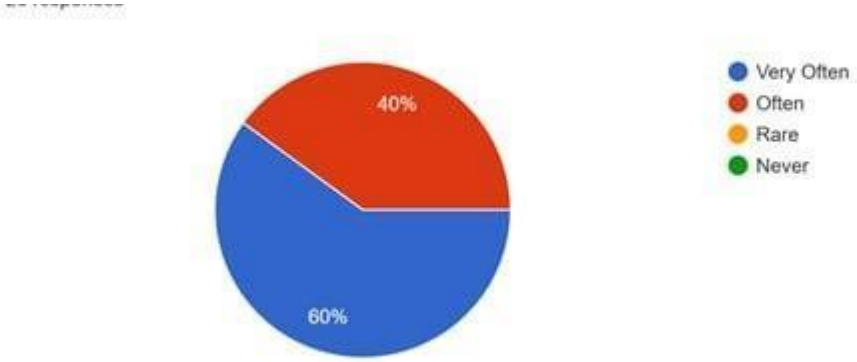


What features do you think are essential in an automobile management system?

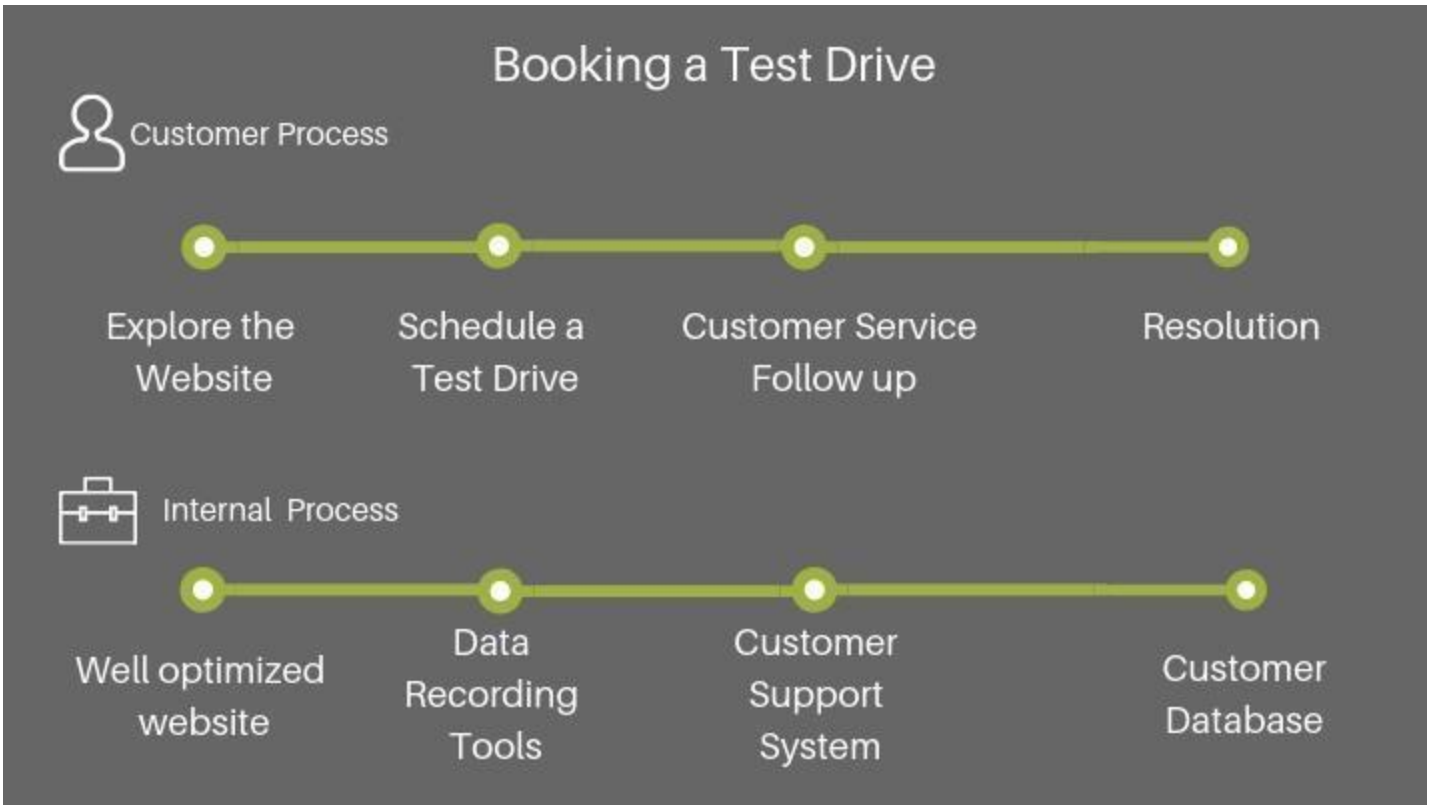
20 responses







1. FIELD STUDY DONE WITH STAKE HOLDER
2. SURVEY ON ONLINE
3. We created A MICROSOFT FORM ON Rental System.



4. THEY GAVE THEIR RESPONSES.

5. ONE OF THE RESPONSE WE ARE ATTACHING HERE

9.2 SOURCE CODE

Admin.jsp

```
<%--

Document : Admin

Created on : 13 Apr, 2021, 12:08:42 PM Author      : KishanVenky

--%>

<%@page contentType=”text/html” pageEncoding=”UTF-8”%>

<!DOCTYPE html>

<html lang=”en”>

<head>

<!—basic 

<meta charset=”utf-8”>

<meta http-equiv=”X-UA-Compatible” content=”IE=edge”>

<!—mobile metas 

<meta name=”viewport” content=”width=device-width, initial-scale=1”>

<meta name=”viewport” content=”initial-scale=1, maximum-scale=1”>

<!—site metas 

<title>Vehicle Maintenance</title>

<meta name=”keywords” content=””>

<meta name=”description” content=””>

<meta name=”author” content=””>

<!—bootstrap css 
```

```
<link rel=”stylesheet” href=”css/bootstrap.min.css”>

<!—style css 

<link rel=”stylesheet” href=”css/style.css”>

<link rel=”stylesheet” href=”table.css”>

<!—Responsive

<link rel=”stylesheet” href=”css/responsive.css”>

<!—fevicon 

<!—Scrollbar Custom CSS 

<link rel=”stylesheet” href=”css/jquery.mCustomScrollbar.min.css”>

</head>

<!—body 

<body class=”main-layout”>

<!—loader 

<div class=”loader_bg”>

</div>

<!—end loader 

<!—header 

<header>

<!—header inner 

<div class=”container-fluid”>


<div class=”row”>

<div class=”col-lg-3 logo_section”>
```

```
<div class=”full”>
<div class=”center-desk”>
<div class=”logo”> <h1>Vehicle Maintenance</h1></div>
</div>
</div>
</div>
<div class=”col-lg-9”>
<div class=”menu-area”>
<div class=”limit-box”>
<nav class=”main-menu”>
<ul class=”menu-area-main”>
<li>
<a href=”index.html”>Home</a>
</li>
<li class=”active”>
<a href=”Admin.jsp”>Admin</a>
</li>
<li>
<a href=”Service.jsp”>Service</a>
</li>

<li>
<a href=”User.jsp”>User</a>
```

```
</li>

</ul>

</nav>

</div>

</div>

</div>

</div>

</div>

<!—end header inner 

</header>

<!—end header 

<!—revolution slider 

<div class=”banner-slider”>

<div class=”container-fluid”>

<div class=”row”>

<div class=”col-md-7”>

<div id=”slider_main” class=”carousel slide” data-ride=”carousel”>

<center>

<h1 style=”color:whitesmoke”>Admin Login</h1>

<form action=”AdminLAction.jsp” method=”post”>

<table>
```

```
<tr><th>UserName</th><td><input type=”text” name=”uname”
required=””></tD></tr>

<tr><th>Password</th><td><input type=”password” name=”pass”
required=””></tD></tr>

<tr><th></th><td><input type=”submit” value=”Login”>

<input type=”Reset” value=”Reset”>

</tD></tr>

</table>

</form>

</center>

</div>

</div>

<div class=”col-md-5”>

<div class=”full slider_cont_section”>

<h4>Vehicle</h4>

<h3>Maintenance</h3>

<p>

Inspecting vehicle engine and mechanical/electrical components to diagnose issues accurately

Inspecting vehicle computer and electronic systems to repair, maintain and upgrade Conducting routine maintenance work aiming to vehicle functionality and longevity

</p>

<div class=”button_section”>

<a href=”about.html”>Read More</a>
```

```
<a href=”contact.html”>Contact Us</a>

</div>

</div>

</div>

</div>

</div>

</div>

<!—end revolution slider

<!—end footer 

<!—Javascript files

<script src=”js/jquery.min.js”></script>

<script src=”js/popper.min.js”></script>

<script src=”js/bootstrap.bundle.min.js”></script>

<script src=”js/jquery-3.0.0.min.js”></script>

<script src=”js/plugin.js”></script>

        <!—Scrollbar Js Files 

<script src=”js/jquery.mCustomScrollbar.concat.min.js”></script>

<script src=”js/custom.js”></script>

</body>

</html>
```

CHAPTER-10

RESULT/ DISCUSSIONS

10.1 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the. Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS:

UNIT TESTING

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately

FUNCTIONAL TEST

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals. Functional testing is centered on the following items:

Valid Input: identified classes of valid input must be accepted.

Invalid Input: identified classes of invalid input must be rejected.

Functions: identified functions must be exercised.

Output: identified classes of application outputs must be exercised.

Systems/Procedures: interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

10.1.1 TEST CASES

Testcase for Login form:

When a user tries to login by submitting an incorrect ID or an incorrect Password then it displays an error message “NOT A VALID USER NAME”.

FUNCTION	ADMIN
EXPECTED RESULTS	Should Validate the user and check his existence in database
ACTUAL RESULTS	Validate the user and checking the user against the database
LOW PRIORITY	NO
HIGH PRIORITY	YES

Testcase for Login form:

When a user tries to login by submitting an incorrect ID or an incorrect Password then it displays an error message “NOT A VALID USER NAME”.

FUNCTION	USER
EXPECTED RESULTS	Should Validate the user and check his existence in database
ACTUAL RESULTS	Validate the user and checking the user against the database
LOW PRIORITY	NO
HIGH PRIORITY	YES

Testcase for Login form:

When a user tries to login by submitting an incorrect ID or an incorrect Password then it displays an error message “NOT A VALID USER NAME”.

FUNCTION	SERVER
EXPECTED RESULTS	Should Validate the user and check his existence in database
ACTUAL RESULTS	Validate the user and checking the user against the database
LOW PRIORITY	NO
HIGH PRIORITY	YES

10.2 OUTPUT SCREENS

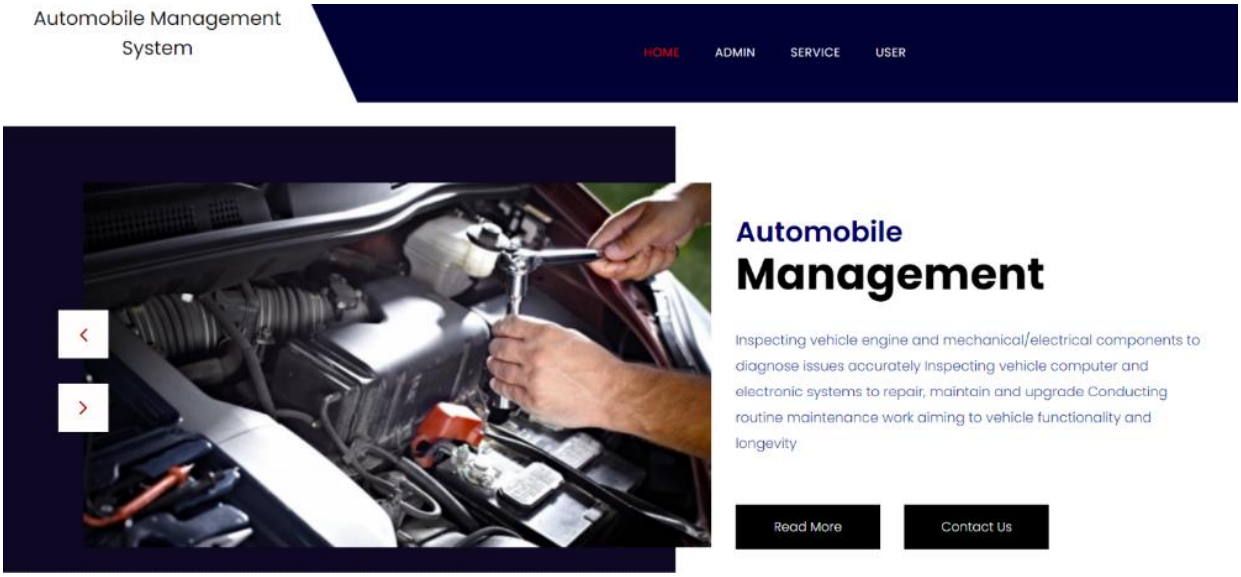


FIG-1

HOME SCREEN

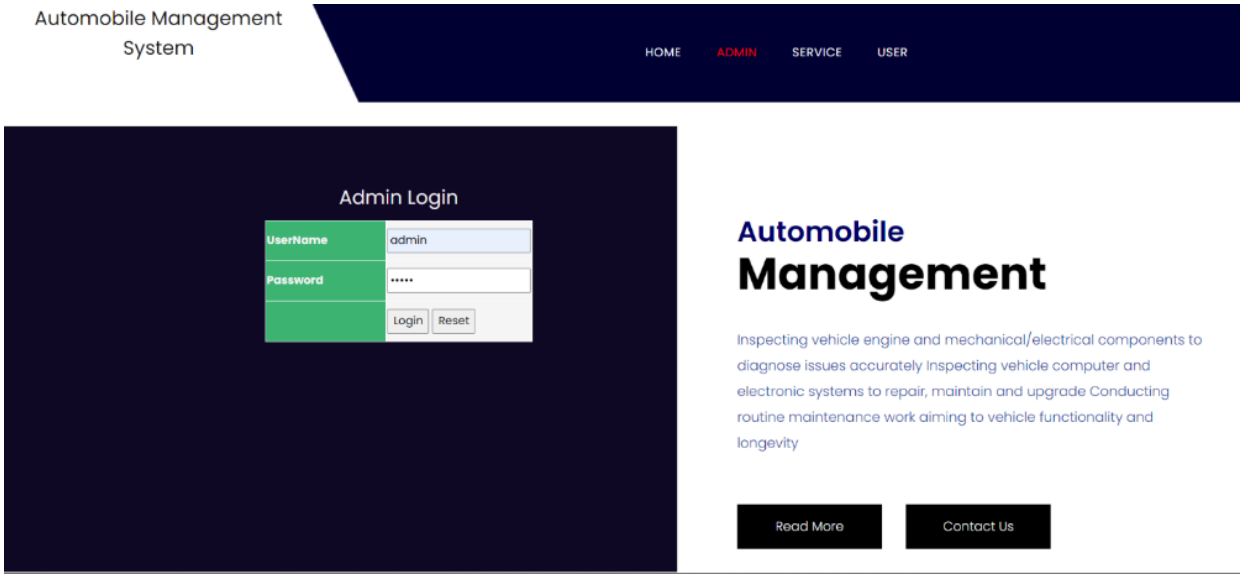


Fig-2

Admin Login Page

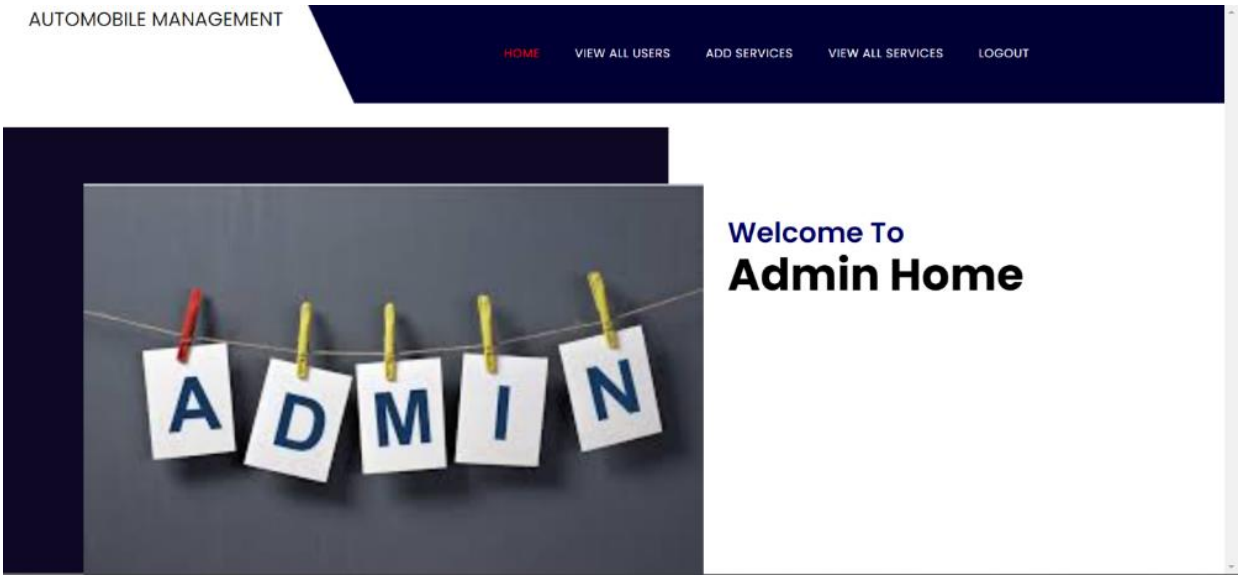

AUTOMOBILE MANAGEMENT
HOME
VIEW ALL USERS
ADD SERVICES
VIEW ALL SERVICES
LOGOUT
ADMIN
Welcome To
Admin Home

Fig-3

ADMIN HOME PAGE

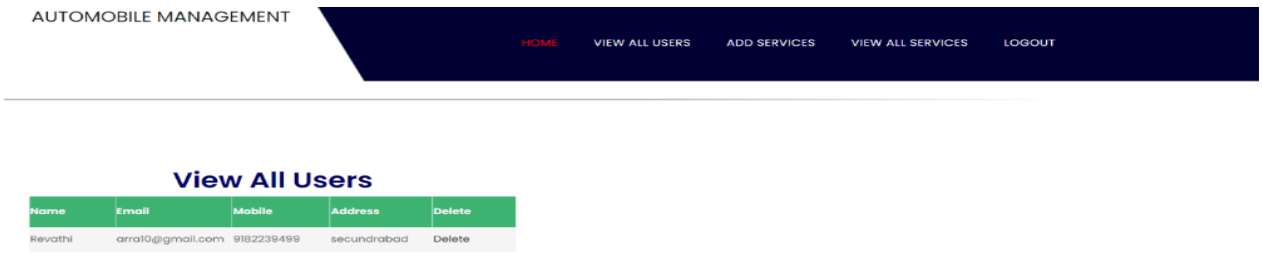


FIG-4

Admin Operation—View Al

FIG-5

ADMIN OPERATIONS—ADD SERVER

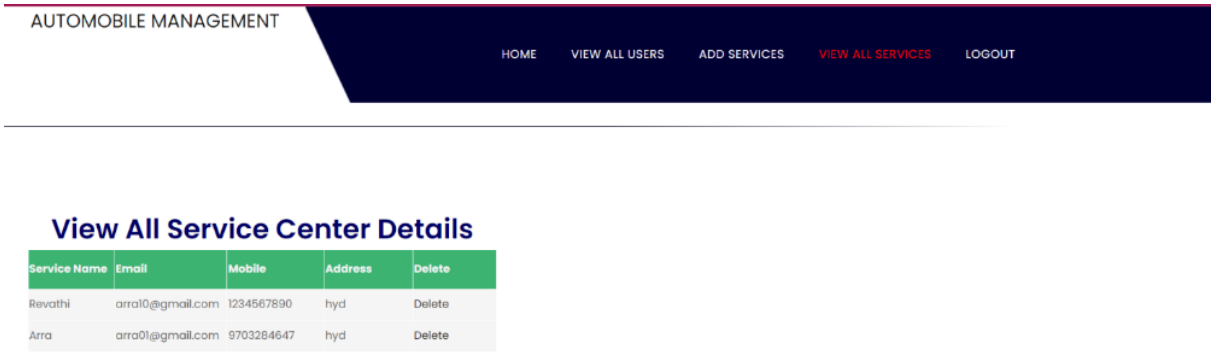
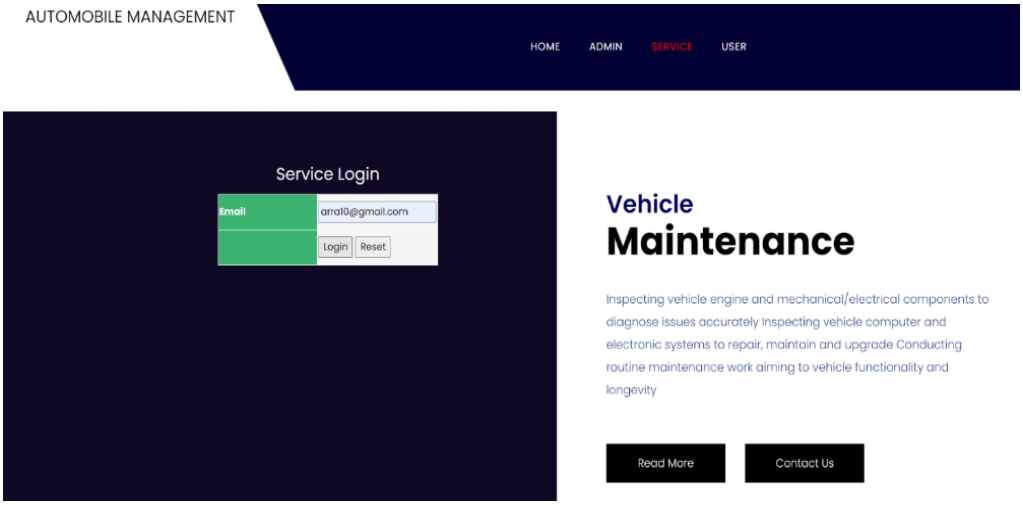


FIG-6

Service Operations Admin Operation------View all Services



Service Login Page

FIG-7

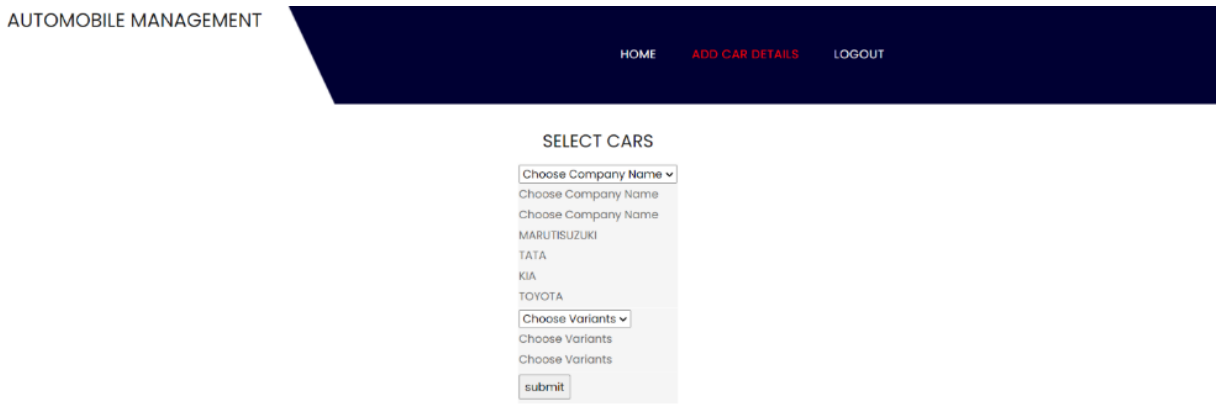

AUTOMOBILE MANAGEMENT
HOME
ADD CAR DETAILS
LOGOUT
SELECT CARS
Choose Company Name
Choose Company Name
Choose Company Name
MARUTISUZUKI
TATA
KIA
TOYOTA
Choose Variants
Choose Variants
Choose Variants
submit

FIG-8

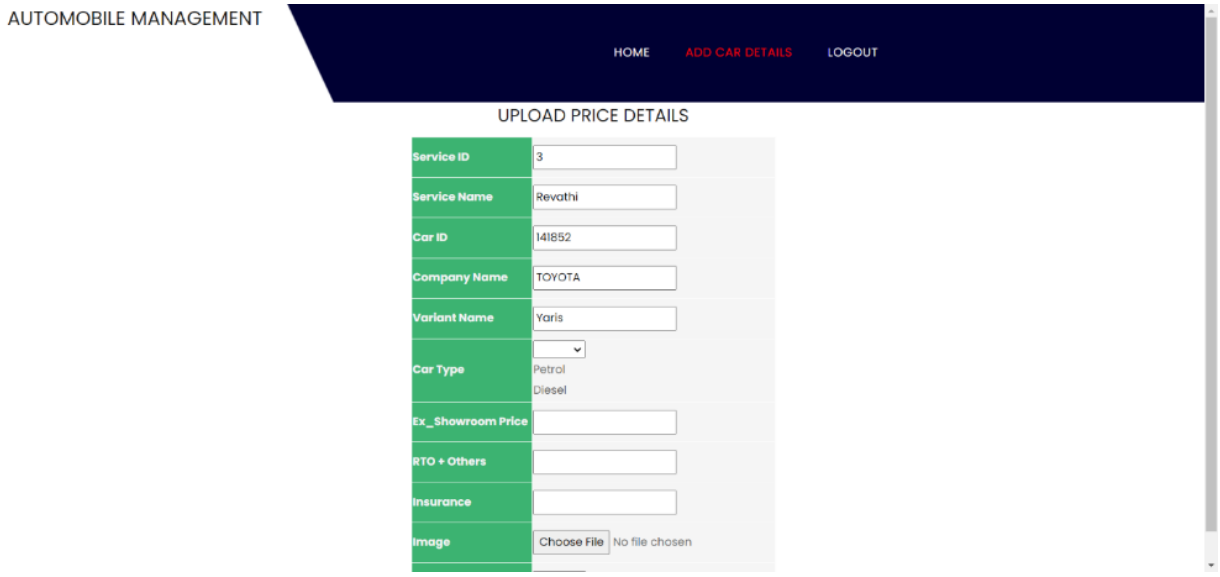

AUTOMOBILE MANAGEMENT
HOME
ADD CAR DETAILS
LOGOUT
UPLOAD PRICE DETAILS
Service ID
3
Service Name
Revathi
Car ID
141852
Company Name
TOYOTA
Variant Name
Yaris
Car Type
Petrol
Diesel
Ex_Showroom Price
RTO + Others
Insurance
Image
Choose File
No file chosen

FIG-9

Service Operations

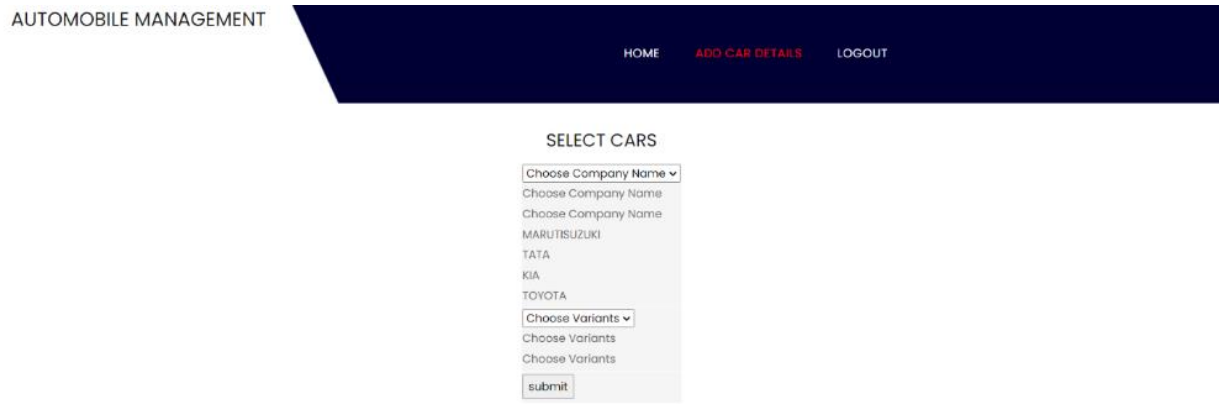


FIG-10

Service Operations

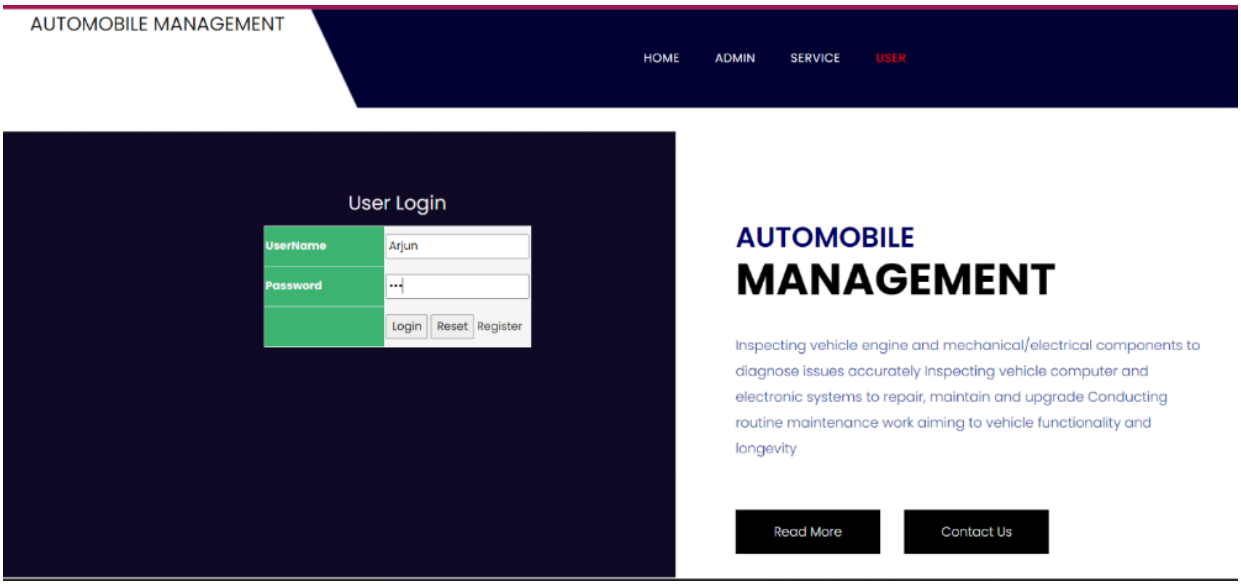


FIG-11

User Login Page

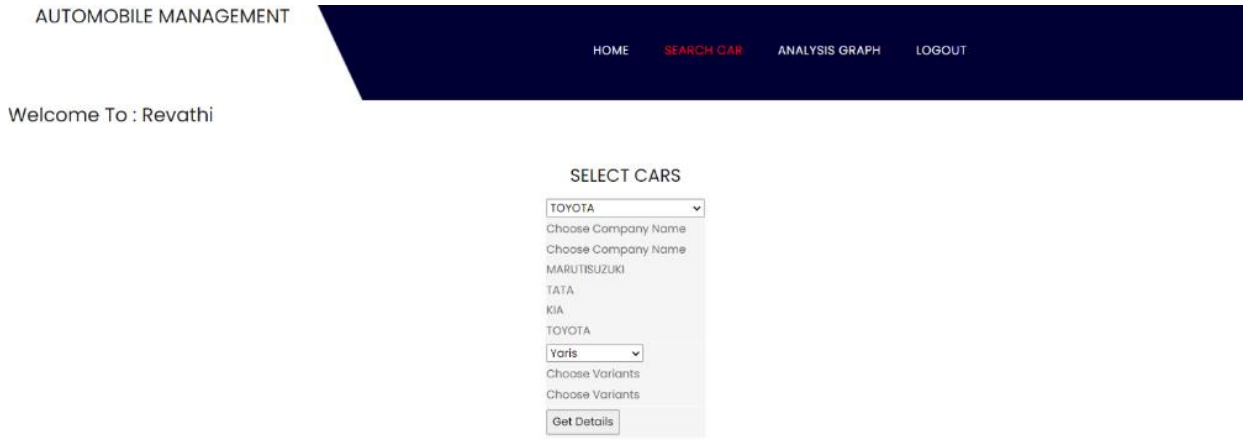


FIG-12

User Operations

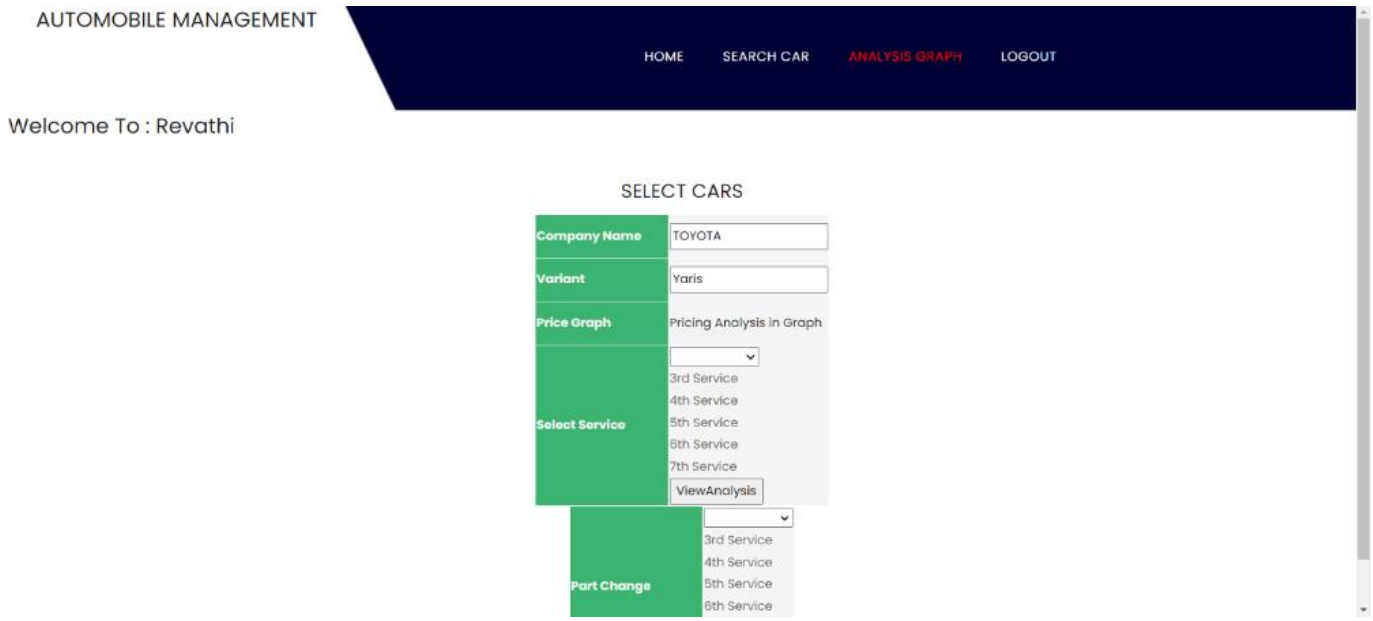


FIG-13

CHAPTER-11

CONCLUSION

11.1 CONCLUSION

The package was designed in such a way that future modifications can be done easily. The following conclusions can be deduced from the development of the project.

- Automation of the entire system improves the efficiency
- It provides a friendly graphical user interface which proves to be better when compared to the existing system.
- It gives appropriate access to the authorized users depending on their permissions.
- It effectively overcomes the delay in communications Updating of information becomes so easier.
- System security, data security and reliability are the striking features.
- The System has adequate scope for modification in future if it is necessary.

We implemented a system which is composed of a combination of a low-cost hardware unit and a user-friendly Android-based mobile application software utilized to create an on-board vehicle diagnostic system. The mobile application software will interact with the hardware interface unit wirelessly via Bluetooth to acquire desired vehicle parameters. These readings will be displayed locally to the user then can be sent to a remote maintenance server as HTTP packets via a cellular internet connection. The packets received will be tabulated in the server, then made use of by the service centre which holds the server.

11.2 FUTURE SCOPE

This application avoids the manual work and the problems concern with it. It is an easy way to obtain the information regarding the various products information that are present in the Super markets. Well, I and my team members have worked hard in order to present an improved website better than the existing one's regarding the information about the various activities. Still, we found out that the project can be done in a better way.

Primarily, when we request information about a particular product it just shows the company, product id, product name and no. of quantities available. So, after getting the information we can get access to the product company website just by a click on the product name.

The next enhancement that we can add the searching option. We can directly search to the particular product company from this site. These are the two enhancements that we could think of at present

CHAPTER-12

REFERENCES

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3. Muhammad Kamal Arif Bin Razali (2017). Thesis Final Year Project University Sultan Zainal Abidin. Online Medical Appointment for Pusat Kesihatan Unisza Using Mobile Application.

The following books were referred during the analysis and execution phase of the project MICROSOFT .NET WITH C# Microsoft .net series

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2. ASP .NET WITH C# 2005 Apress Publications

3. C# COOK BOOK O reilly Publications

4. PROGRAMMING MICROSOFT ASP .NET 2.0 APPLICATION Wrox Professional Guide

5. BEGINNING ASP .NET 2.0 E-COMMERCE IN C# 2005 Novice to Professional.

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[5] L Ferhatovic, A. Lipjankic. A. Handzic, and N. Nosovic, “System for remote diagnostic of vehicle defects,” in Proc. 17th Telecommunications Forum (TELFOR), pp. 1323-1326, November 2009